ST. EDMUND'S SECONDARY SCHOOL

INFORMATION AND COMMUNICATION TECHNOLOGY (ICT)

PRODUCTIVITY TOOL SPREADSHEETS

FORM 2

HAKUBEZA CLETUS (Mr.)



Overview

Today, we are going to learn about Productivity Tools with a focus on Microsoft Excel (Spreadsheet).

Microsoft Excel is a computer program used to organize, calculate, and present information using rows and columns called spreadsheets. It helps people perform tasks faster and more accurately in schools, businesses, hospitals, banks, and many other places.

In this topic, you will understand the importance of spreadsheets, identify the main parts of the Excel window, enter and format data, and use simple formulas to solve everyday problems.

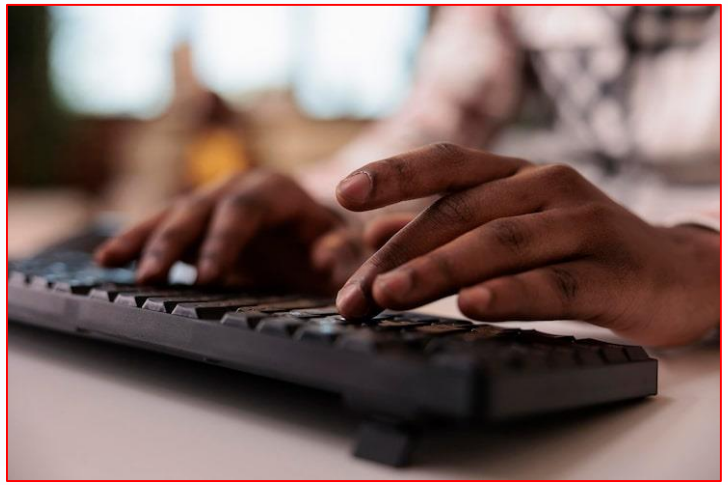


Specific Competencies

2.2.2.1 *Manage data using spreadsheets*

Expected Standard

✓ ***Data managed using spreadsheets***



Discuss

- What is a spreadsheet, and where have you seen it being used in daily life?
- Why do you think people use Microsoft Excel instead of calculating everything manually?

Introduction

In this lesson, you will explore how Spreadsheet can be used as a productivity tool to organize and analyze information.

You will also practice using spreadsheet functions such as **MIN**, **MAX**, **COUNT**, **COUNTA**, and **SUM-IF** to perform calculations and make work easier and faster.

You will create a simple budget using a spreadsheet to record income and expenses.

In addition, you will improve the appearance and clarity of their work by adding borders, colors, and bold text to format the spreadsheet neatly and professionally.



MS WORD vs MS EXCEL: KEY DIFFERENCES

MS WORD (WORD PROCESSOR)

- **CREATING DOCUMENTS** (reports, letters, resumes)
- **TEXT-BASED CONTENT** Typing, formatting paragraphs)
- **PAGE LAYOUTS & DESIGN** margins, headers, styles, images
- **EDITING TOOLS** spell check, grammar, changes
- **EXPORT AS PDF/DOCX**
- Focus on writing and readability

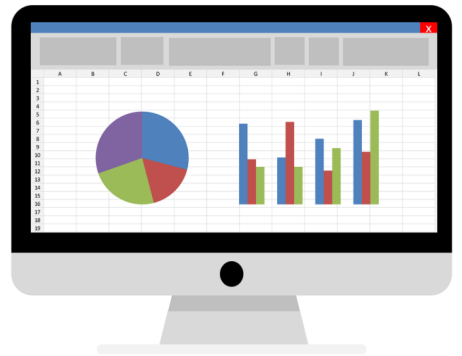
VS

MS EXCEL (SPREADSHEET)

- **DATA MANAGEMENT** Organizing data in grids)
- **NUMERICAL CALCULATIONS** (formulas, functions, analysis)
- **CHARTS & GRAPHS** Visualizing data and trends
- **SORTING & FILTERING** manipulating datasets
- **EXPORT AS XLSX/CSV**
- Focus on calculation and analysis

Some common uses of Spreadsheets

- **Preparing budgets:** used to record income and expenses.
- **Performing calculations:** helps users add, subtract, multiply, and divide quickly using formulas and functions.
- **Keeping records:** stores information such as marks, attendance, sales, or stock.
- **Creating tables and charts:** presents information clearly using graphs and charts.
- **Organizing information:** arranges data neatly in rows and columns for easy understanding.



Introduction to Microsoft Excel

Microsoft Excel is a computer program used to **ente**r, **analyze** and present quantitative data.

A **spreadsheet** is a grid of rows and columns in which you enter text, numbers, and the results of calculations

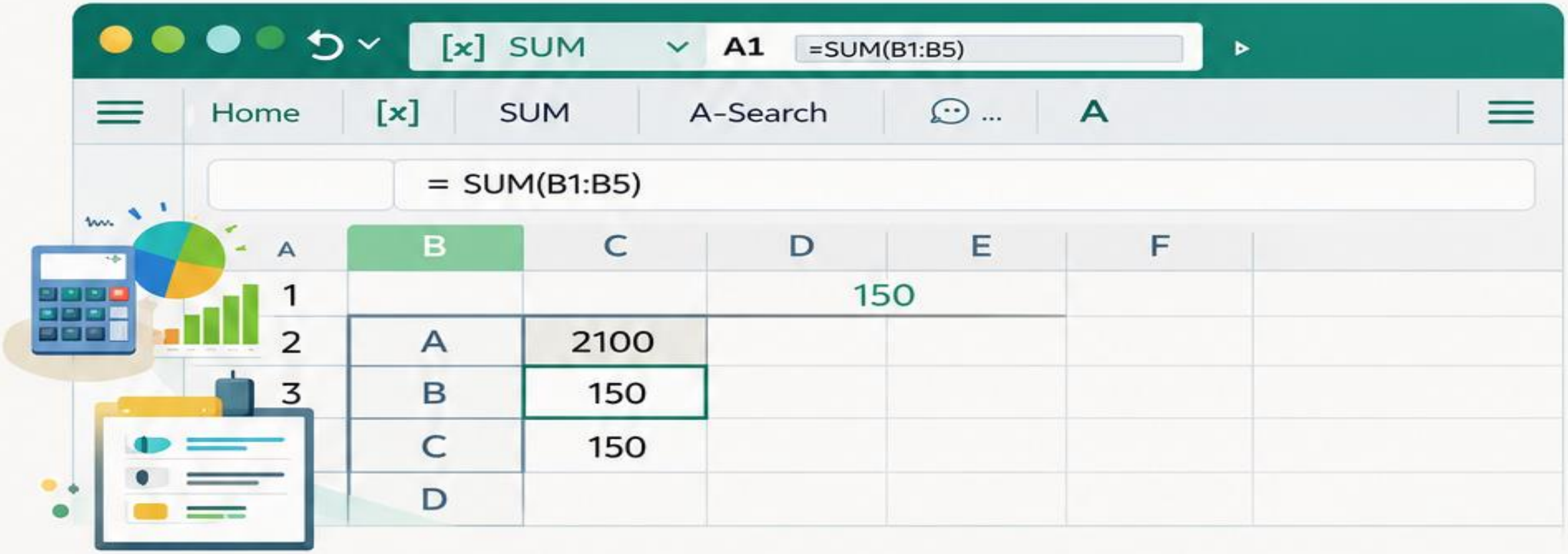


It is a database created by Microsoft which is applicable in different operating systems like windows.

It offers the feature of numbered **rows** and **columns** which can be used for organizing data in a systematic manner.

What is Spreadsheet Software?

A **spreadsheet** is a software used to organize, analyze, and manipulate data, using a grid of rows and columns.



Introducing Spreadsheets

It is a single page in Excel where data is entered and calculations are done. It is also called a worksheet.

The file used to store worksheets is called a workbook.

Workbook: A workbook is the entire Excel file that contains one or more spreadsheets (worksheets).

Common spreadsheet programs include **Microsoft Excel**, **Google Sheets**, and **LibreOffice Calc**.



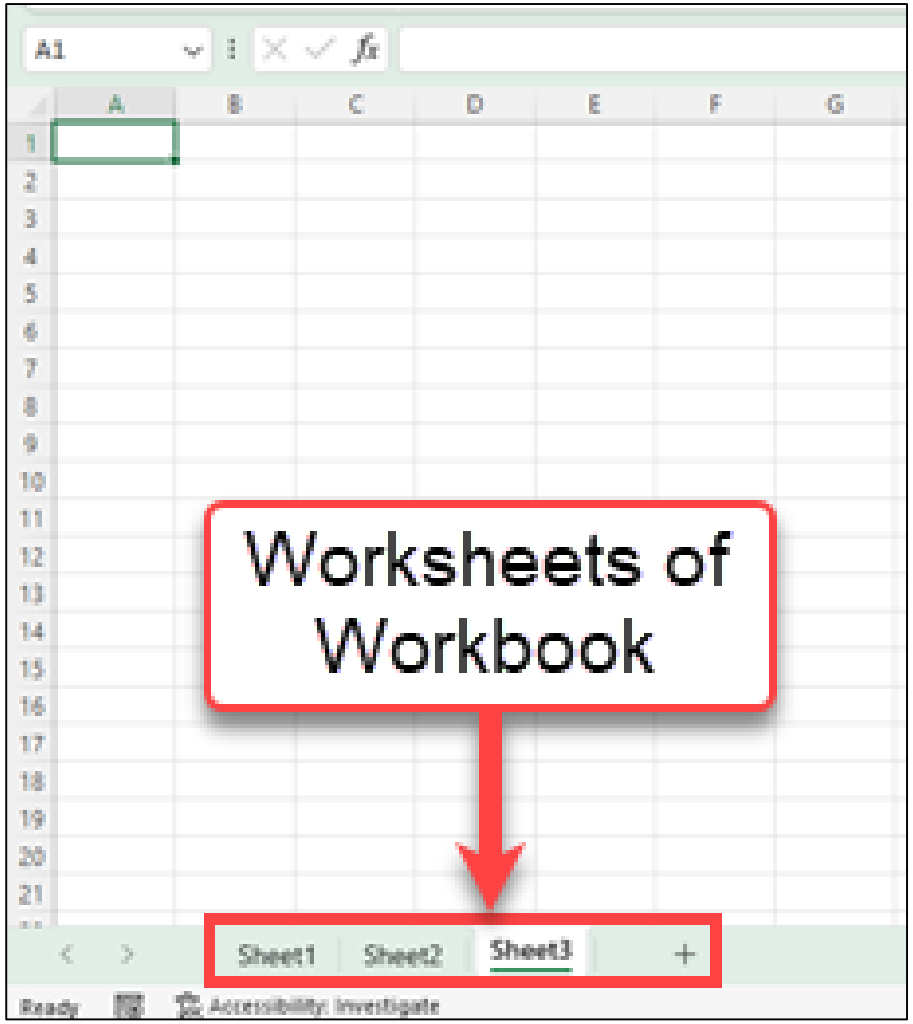


Example:

If you save a file called ***"School Budget.xlsx"***, that file is the workbook. Inside it, you may have different spreadsheets such as:

- January Budget
- February Budget
- March Budget

Each page is a **spreadsheet/worksheet**, while the whole file is the workbook.



KEY FEATURES

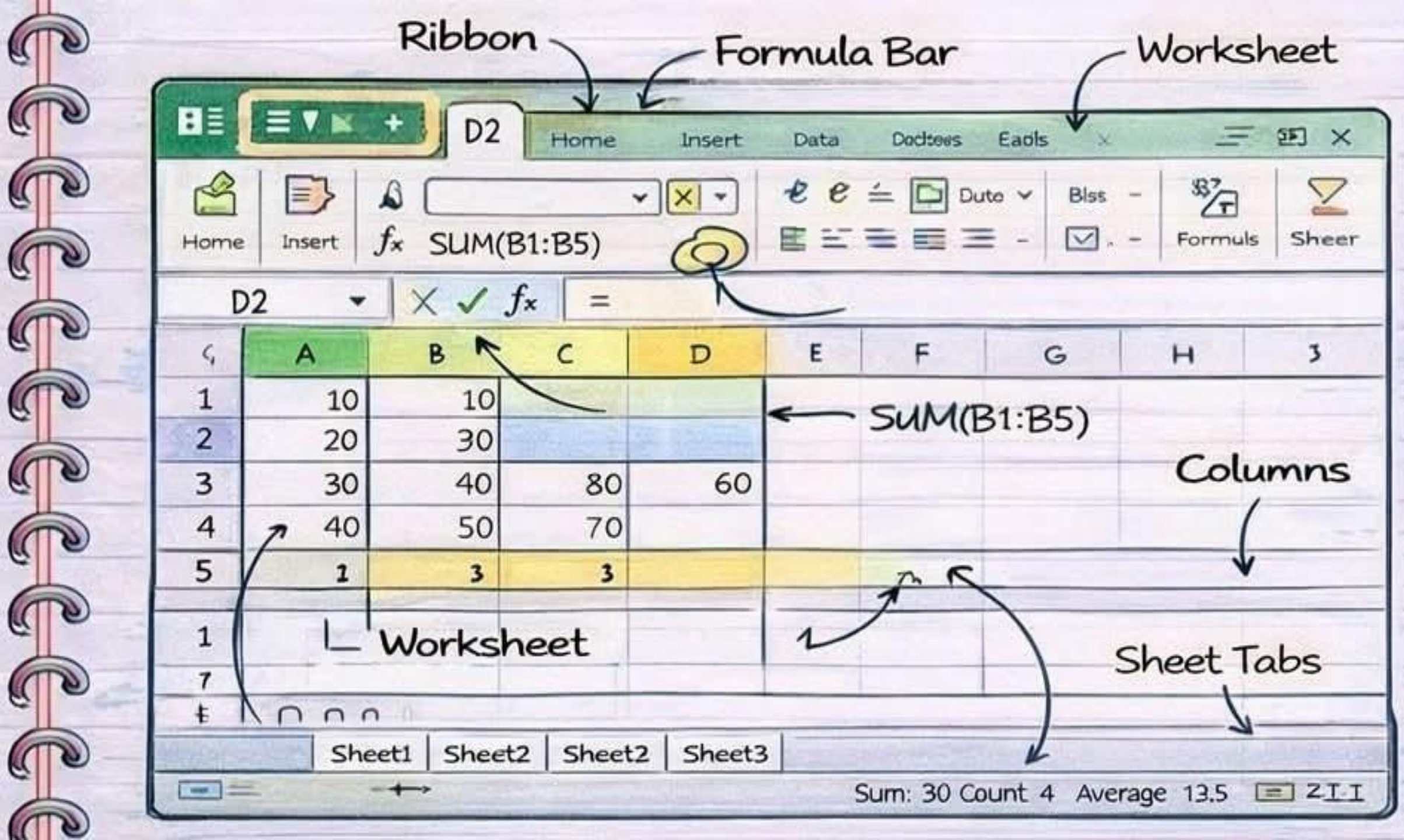






Here's a very brief and beginner-friendly overview of the key Excel features and tools:

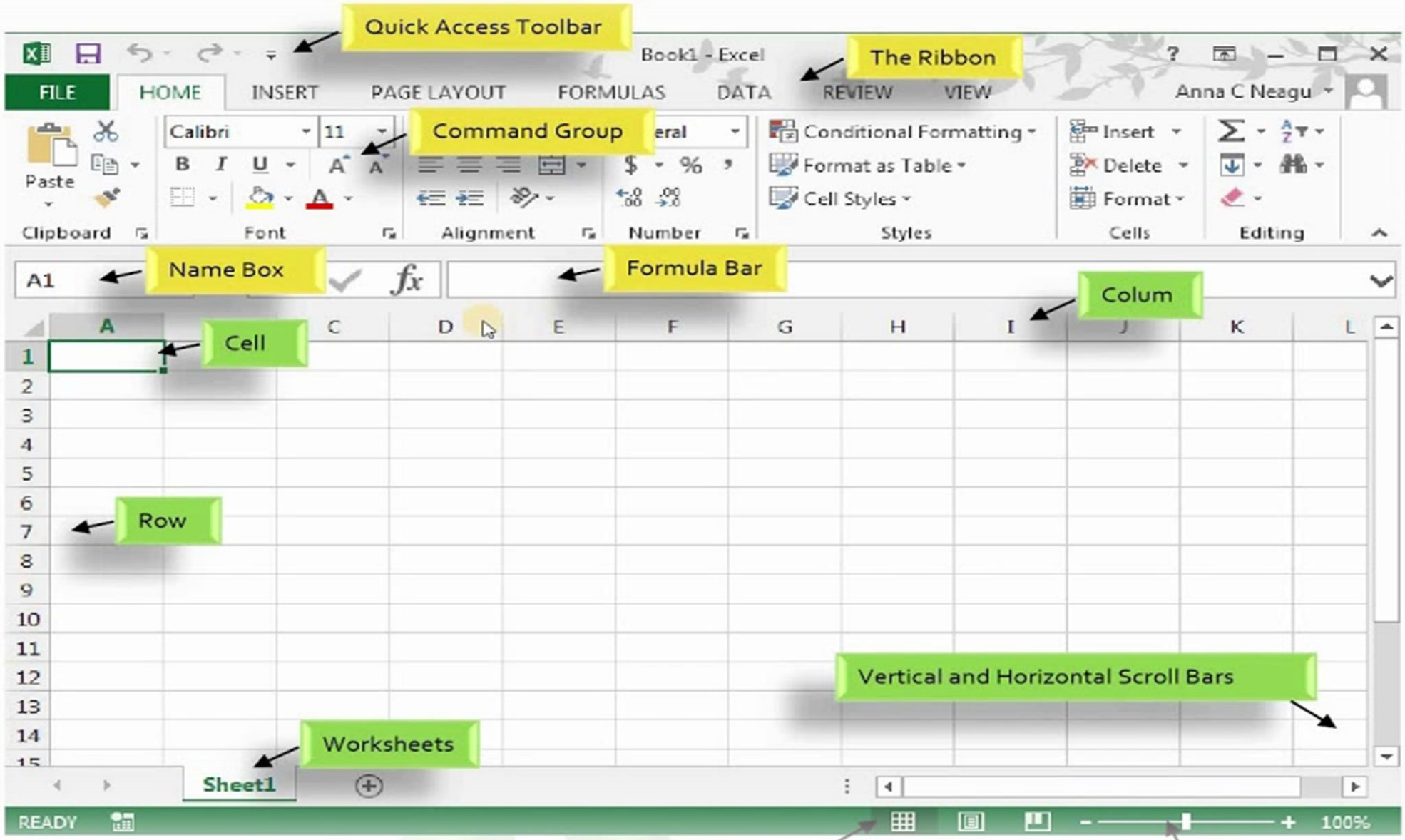
- ✓ **Workbook** – The Excel file that contains one or more worksheets.
- ✓ **Worksheet** – A single sheet where data is entered and analyzed.
- ✓ **Rows** – Horizontal lines identified by numbers (1, 2, 3...).
- ✓ **Columns** - Vertical lines identified by letters (A, B, C...).
- ✓ **Cell** - The intersection of a row and a column where data is entered.
- ✓ **Cell Example: A1** – The location of a cell (e.g, A1).


Ribbon
Formula Bar
Worksheet
D2
Home
Insert
Data
Home
Insert
SUM(B1:B5)
Formuls
Sheer
D2
=
A
B
C
D
E
F
G
H
1
10
10
2
20
30
3
30
40
80
60
4
40
50
70
5
1
3
3
SUM(B1:B5)
Columns
Worksheet
Sheet Tabs
Sheet1
Sheet2
Sheet2
Sheet3
Sum: 30 Count 4 Average 13.5

- **Workbook** – The Excel file that contains one or more worksheets.
- **Worksheet** – A single sheet where data is entered and analyzed.
- **Rows** – Horizontal lines identified by numbers (1, 2, 3...).
- **Columns** – Vertical lines identified by letters (A, B, C...).
- **Cell :** A1 - The location of a cell (e.g. A1).
- **Range** – A group of selected cells. Eg (A1:D1).
- **Formula Bar** – Displays and allows editing of data or formulas in a cell.
- **Ribbon** – The toolbar that contains Excel commands and tools.
- **Name Box** – Shows the address or name of the selected cell.
- **Sheet Tabs** – Used to switch between worksheets.
- **Status Bar** - Shows basic calculations like sum or count of selected cells.

Sheet Tabs

file:///C:/Users/AZEEZ/Documents/Microsoft%20Excel%20Book%20pdf/Excel%20Training%20-%20Level%201.pdf



Spreadsheet…cont

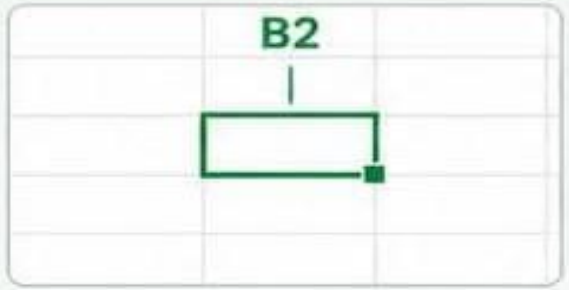
A Spreadsheet document contains **rows** and **columns** that holds data and formula. There are ***16,384*** columns and ***10,48,576*** rows in each worksheet.

- The columns are named **A,B,C** …….up to **Z**.
- Rows are numbered from **1** to ***10,48,576***.
- A box (column-row intersection) in a worksheet is called a **cell**. There are ***17,179,869,184*** cells in each worksheet.

Each cell has a cell address or cell reference. A cell address is formed by column letter and row number.

Example: **B2** is a cell address for the cell located in **column B** and **row 2**.

1

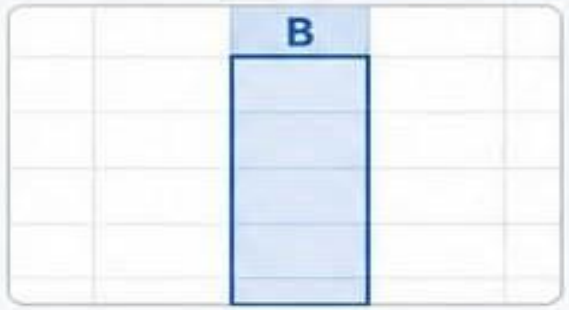


Cells & Rows

A cell is the intersection of a row and a column, where data is entered. Rows run horizontally and are identified by numbers (1, 2, 3, ...).

Example:
B2
Column B, Row 2

2



Columns

Columns run vertically from top to bottom and are identified by letters (A, B, C, ...).

Example:
Column B
Vertical range of cells

3



Worksheets

A **worksheet** is a single spreadsheet within a workbook. You can have multiple worksheets to organize your data.

Example:
Sheet1
One worksheet tab

4

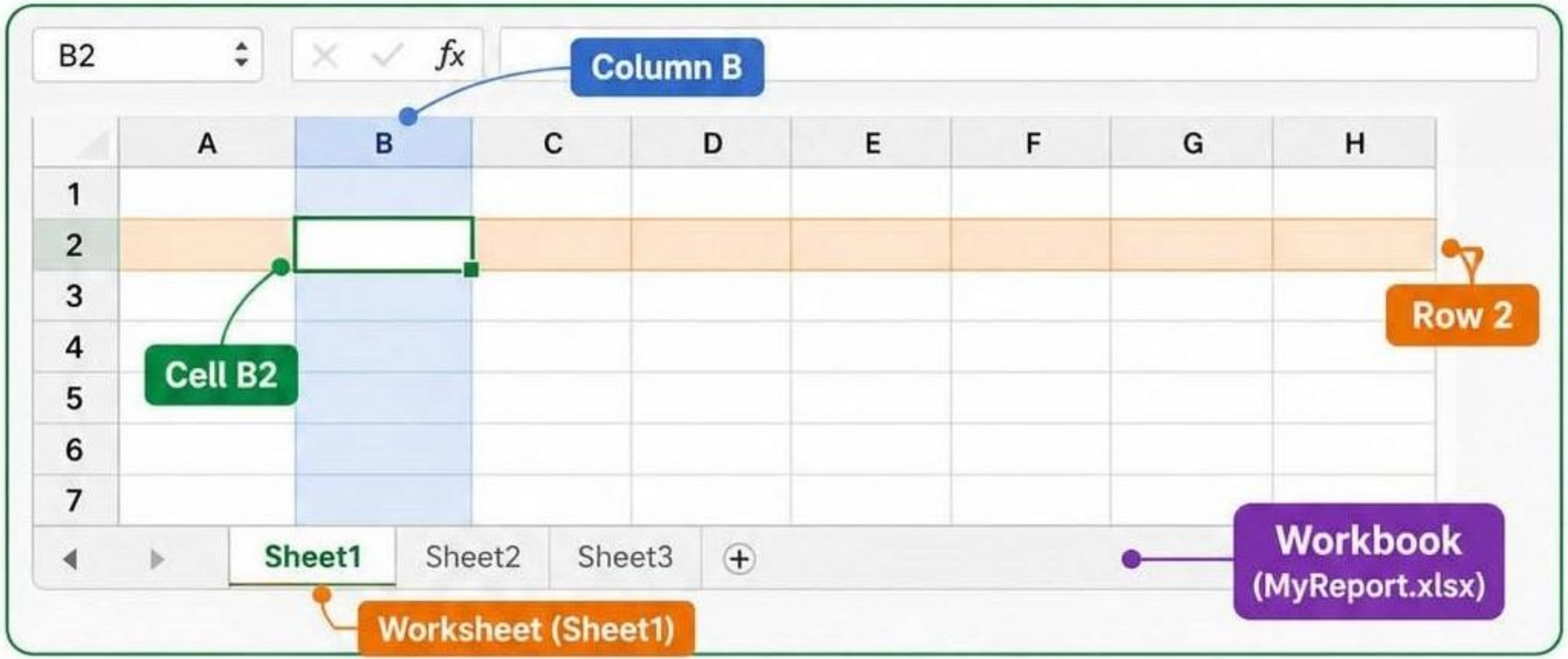


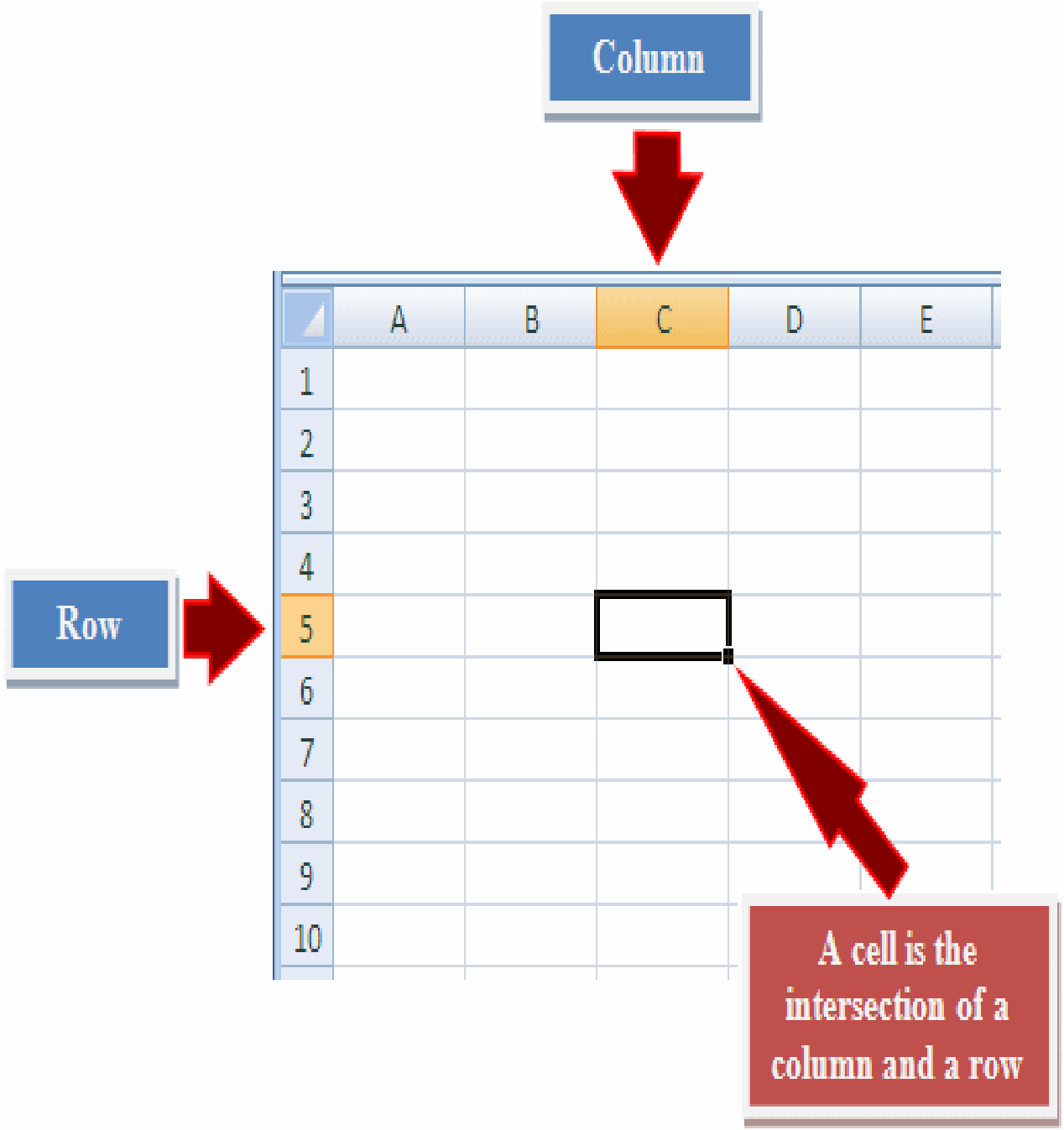
Workbook

A **workbook** is the entire Excel file that contains one or more **worksheets**. It is saved with the .xlsx extension.

Example:
MyReport.xlsx
The entire Excel file

Spreadsheets on a Workbook

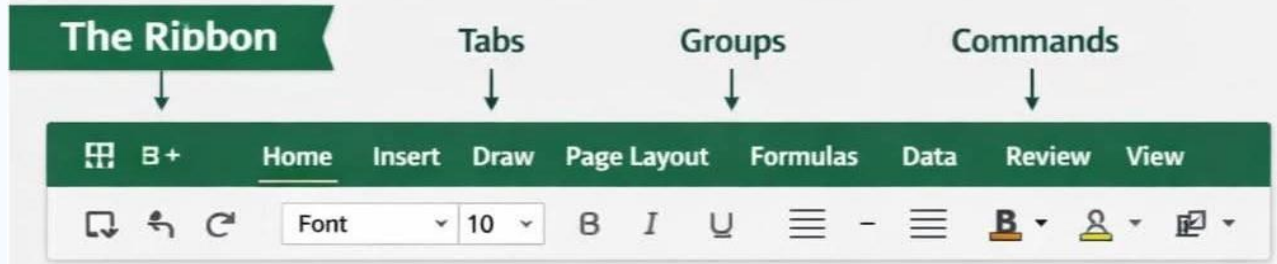




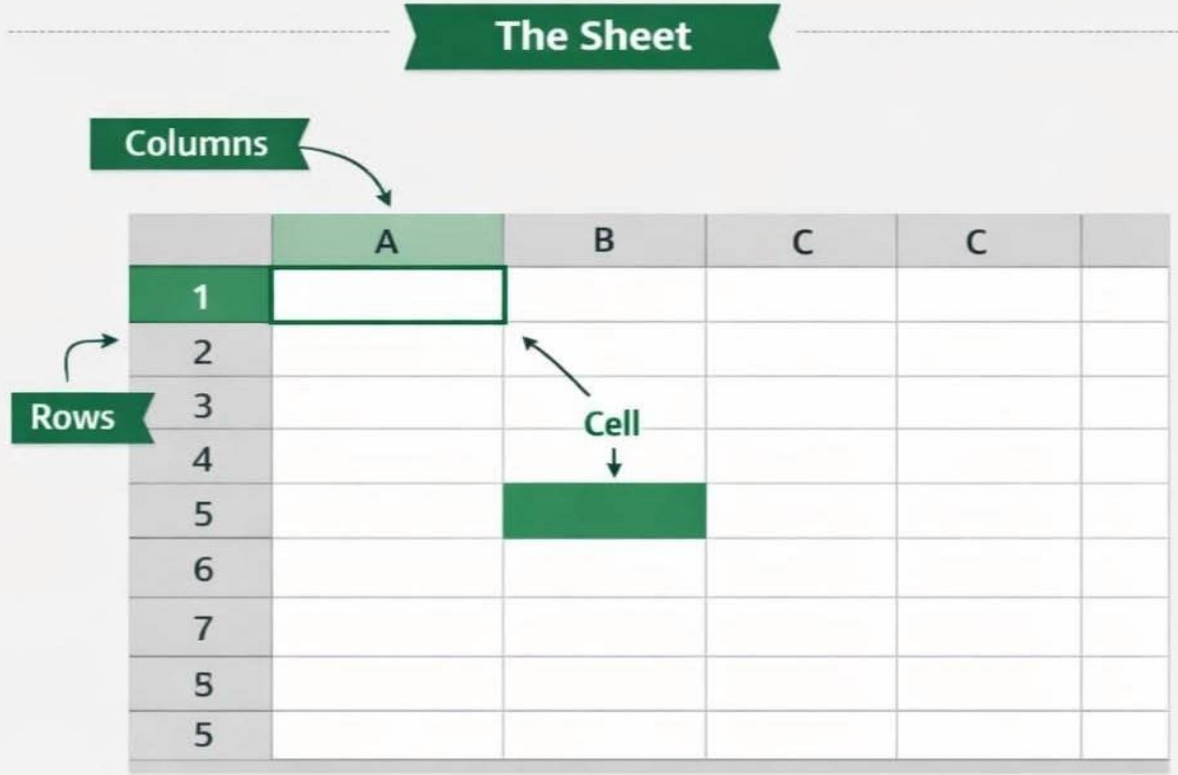
Cells, Rows, and Columns:

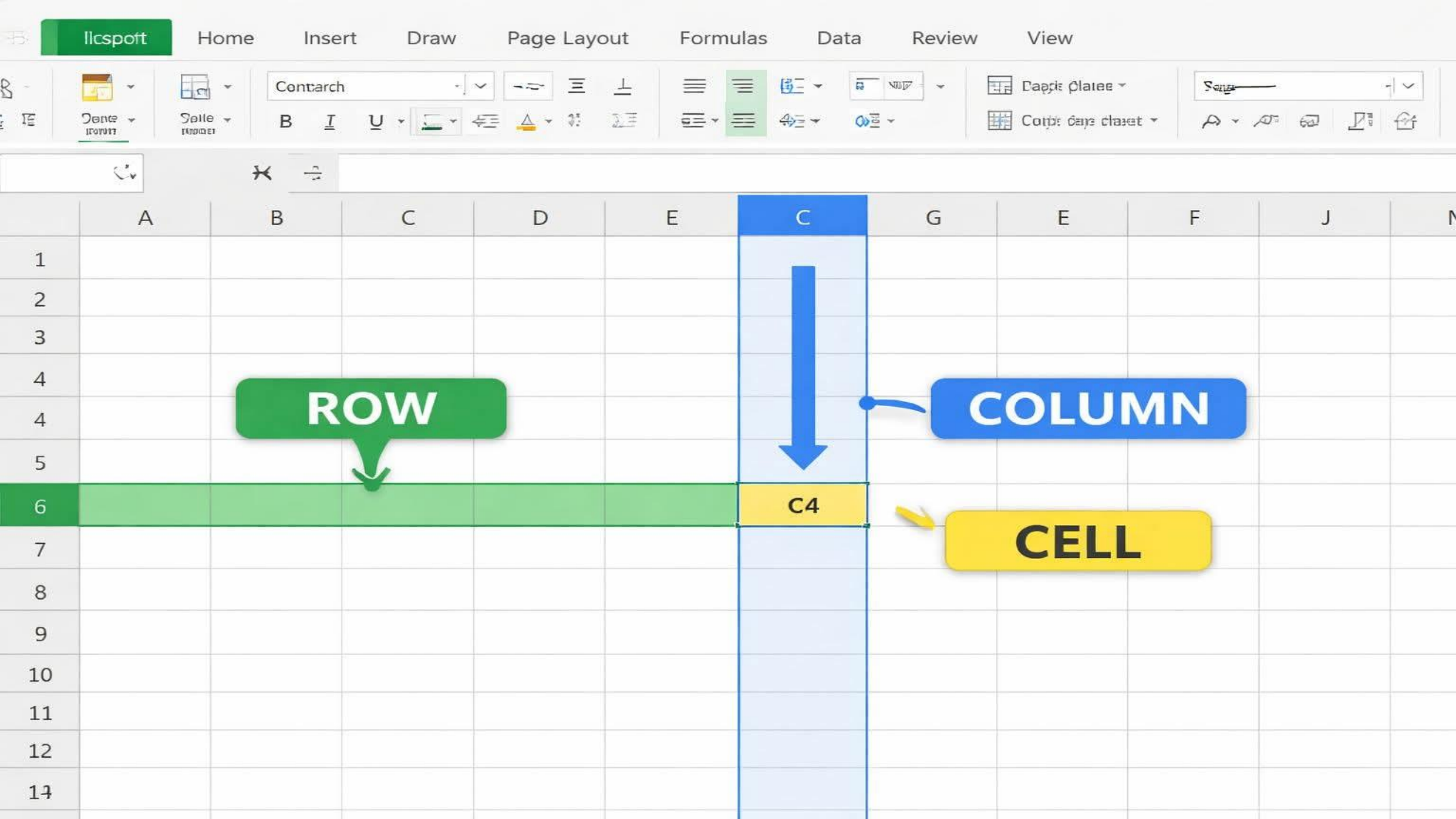
Spreadsheets are divided into cells, which are organized in rows (horizontal) and columns (vertical).

Each cell has a unique address (e.g., A1, B2) to locate data easily.


The Ribbon
Tabs
Groups
Commands
Home
Insert
Draw
Page Layout
Formulas
Data
Review
View
Font
10


Name Box
C4
Cell Reference
C4

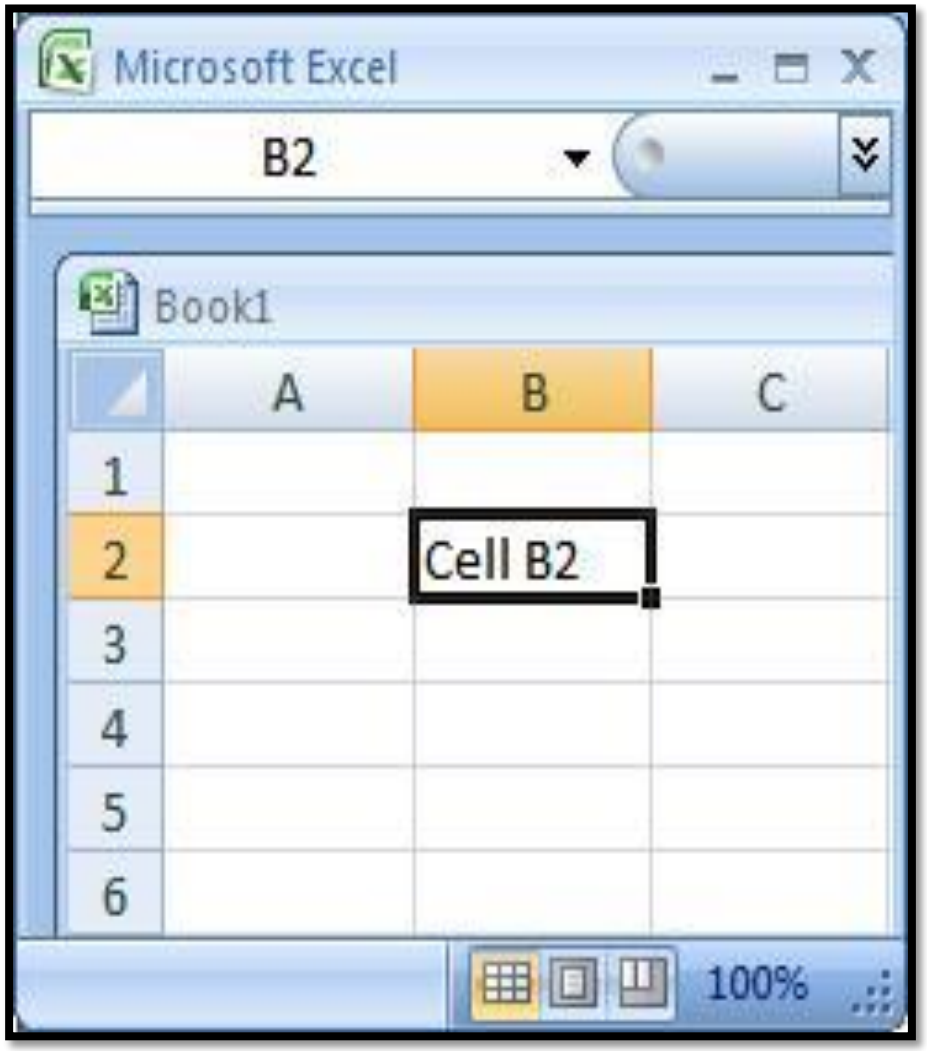

The Sheet
Columns
A
B
C
C
1
2
3
4
5
6
7
5
5
Rows
Cell


Home
Insert
Draw
Page Layout
Formulas
Data
Review
View
A
B
C
D
E
C
G
E
F
J
1
2
3
4
4
5
6
7
8
9
10
11
12
ROW
COLUMN
C4
CELL

A Cell

A cell is the intersection between a column and a row.

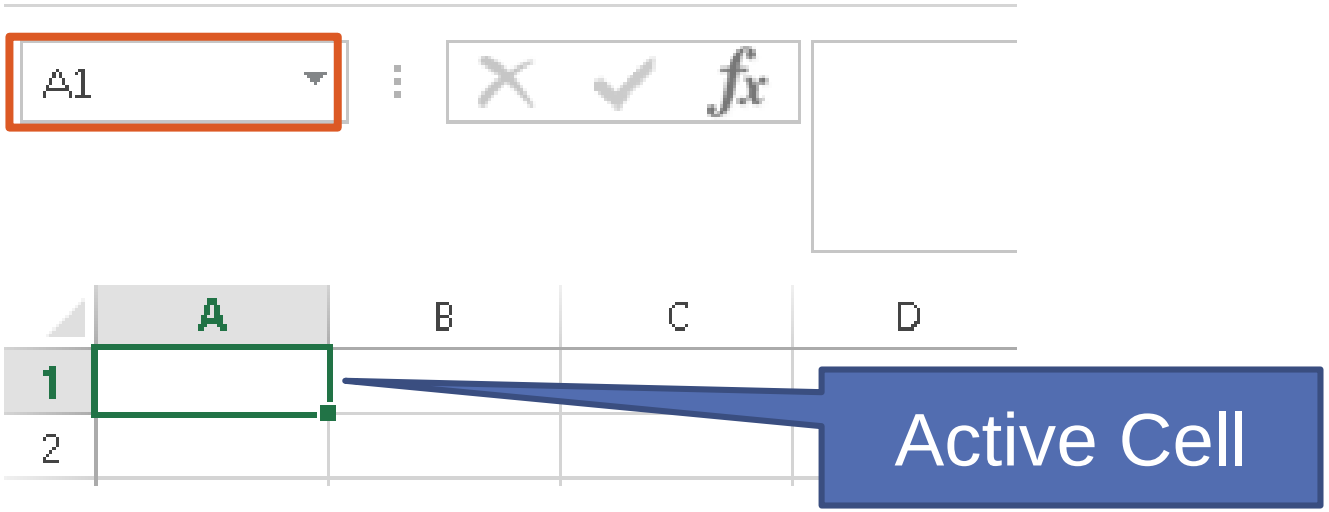
Each cell is named for the column letter and row number that intersect to make it.

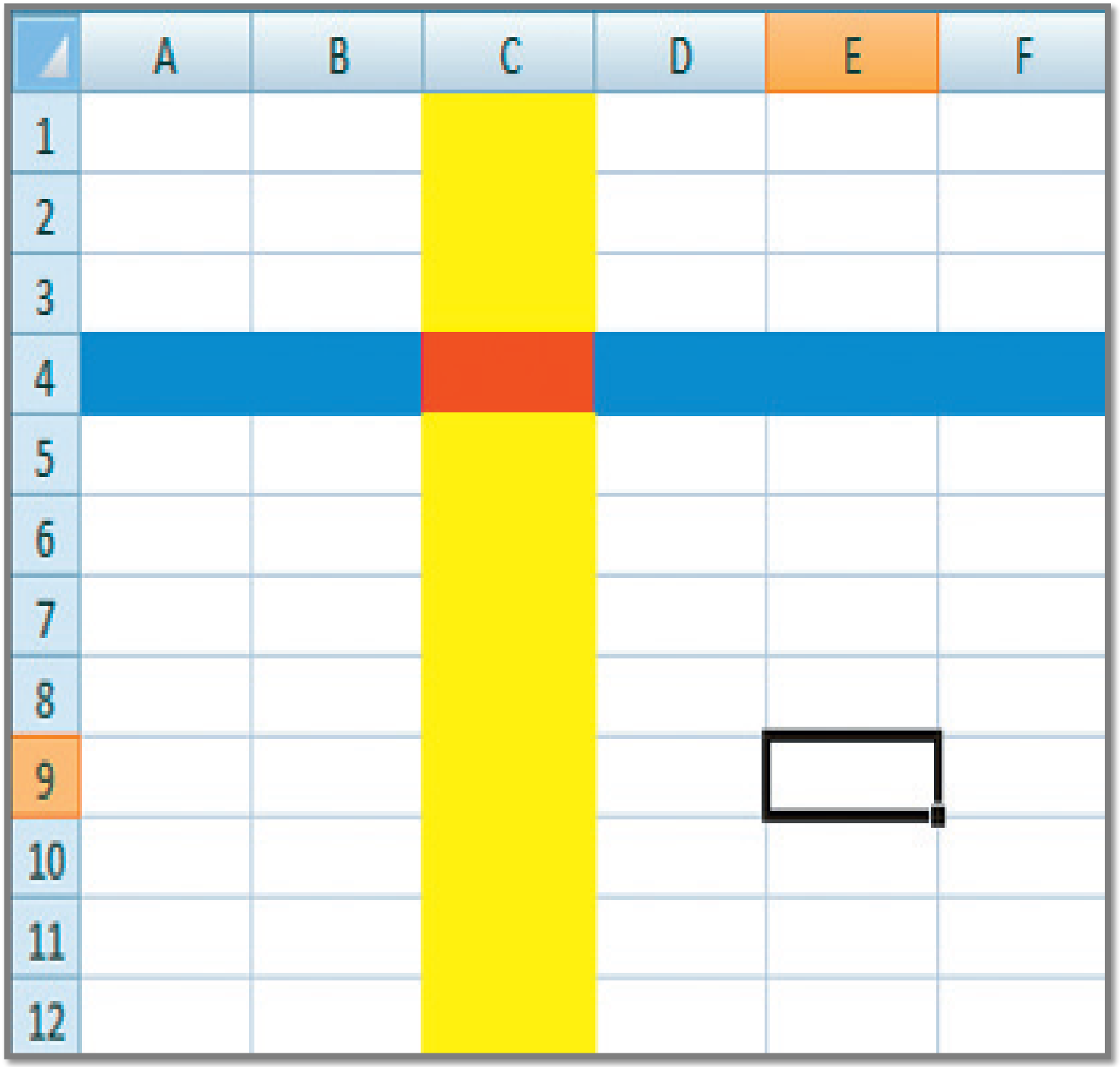


Active Cell

Each cell on the spreadsheet has an address to be referred. This is made up with a column position and row position.

e.g. A1 means the cell is located in column A and row 1.





Individual Task

1. Identify the following using colors shaded in the spreadsheet.
- Row
- Column

2. Mention the cell address shaded in red.

3. What is a cell reference for the active cell?

4. Explain the difference between a spreadsheet and a Workbook.

Activity 1: Exploring Spreadsheets in Daily Life

Part A: Scenario (Think & Discuss)

Your school is organizing a Sports Day event. The class teacher asks you to help prepare information using a spreadsheet. You need to record:

- Names of learners
- Amount of money contributed
- Items needed for the event

1. Explain how spreadsheet can be helpful in this situation.
2. Mention 4 features that are found on the Microsoft Excel Window.
3. Write a file extension of an Excel document.

Part B: Questions (Group Work)

1. Give three uses of spreadsheets in daily life.
2. In your own words, explain what a spreadsheet is.
3. What is the name of the area where you type commands like Save and Open?
4. How can spreadsheets help your family or school in managing information more easily?
5. How can colours, borders, and bold text help make a spreadsheet easier to read and understand?
6. Why is a spreadsheet better than writing records in a notebook?

Part C: Practical Task (Hands-on Activity)

- Open Excel and:

1. Enter a list of 5 classmates' names in **column A**
2. Enter their contributions in **column B**

- Identify and label at least:

1. 2 rows
2. 2 columns
3. 3 cells

- Point out the **File tab**, **Ribbon**, **scroll bars**, and **sheet tabs**

Managing data using spreadsheets

Managing data means:

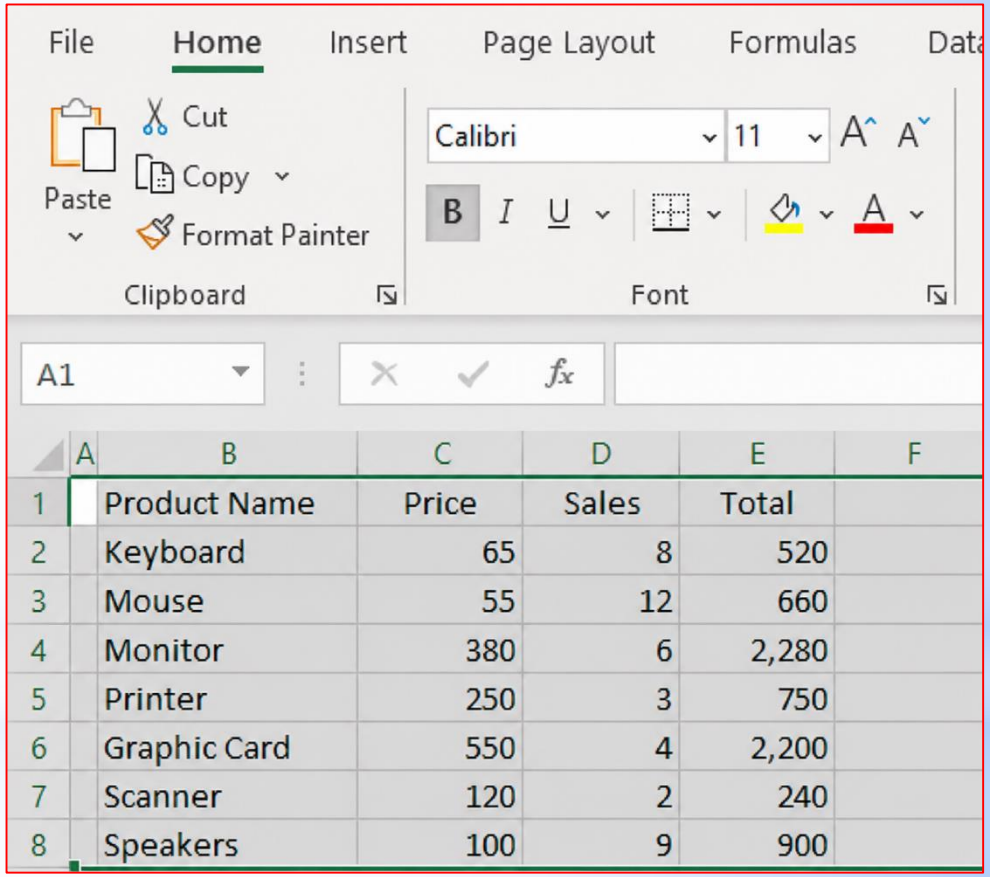
- Entering information correctly
- Organizing information neatly
- Editing or updating information
- Saving and retrieving information easily



A spreadsheet helps us manage data quickly and accurately.

Entering Data in a Cell

- Spreadsheet cells can contain text, numbers, or formulas.
- Text is any combination of letters and numbers and symbols.
- Numbers are values, dates, or times.
- Formulas are equations that calculate a value.
- You enter data in the active cell.



	A	B	C	D	E	F
1		Product Name	Price	Sales	Total	
2		Keyboard	65	8	520	
3		Mouse	55	12	660	
4		Monitor	380	6	2,280	
5		Printer	250	3	750	
6		Graphic Card	550	4	2,200	
7		Scanner	120	2	240	
8		Speakers	100	9	900	

A spreadsheet stores information inside cells

Different types of data entered include:

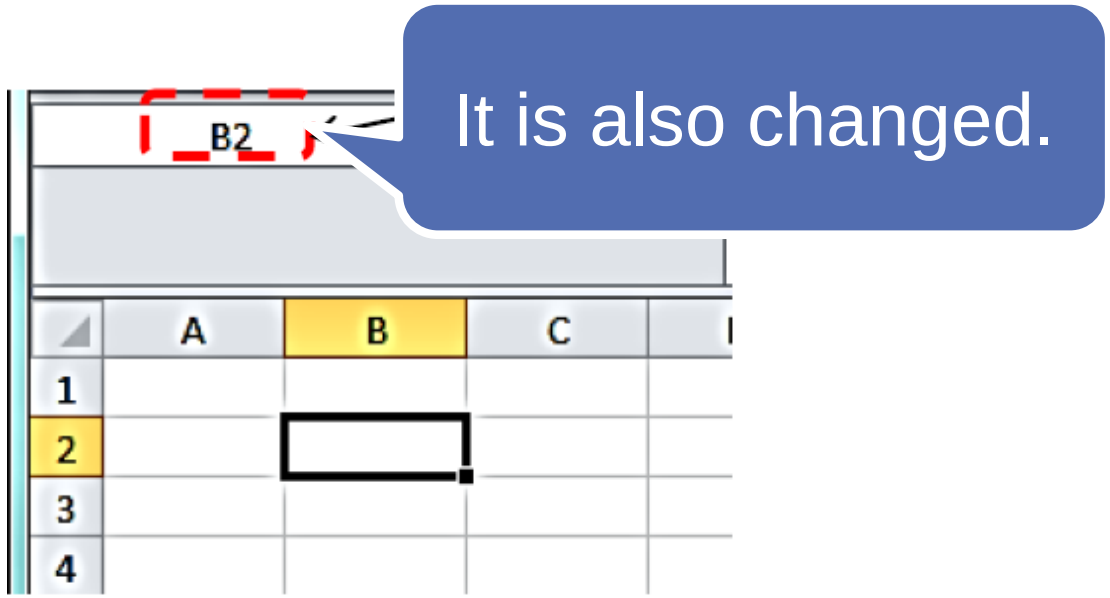
Type of Data	Example
Text/Labels	Names, subjects
Numbers/Values	45, 100, 250
Dates	12/05/2026
Formula	=A1+B1

ENTERING A DATA IN A CELL

1. Select the cell

Before you enter data, you need to select the cell in which you want to put data.

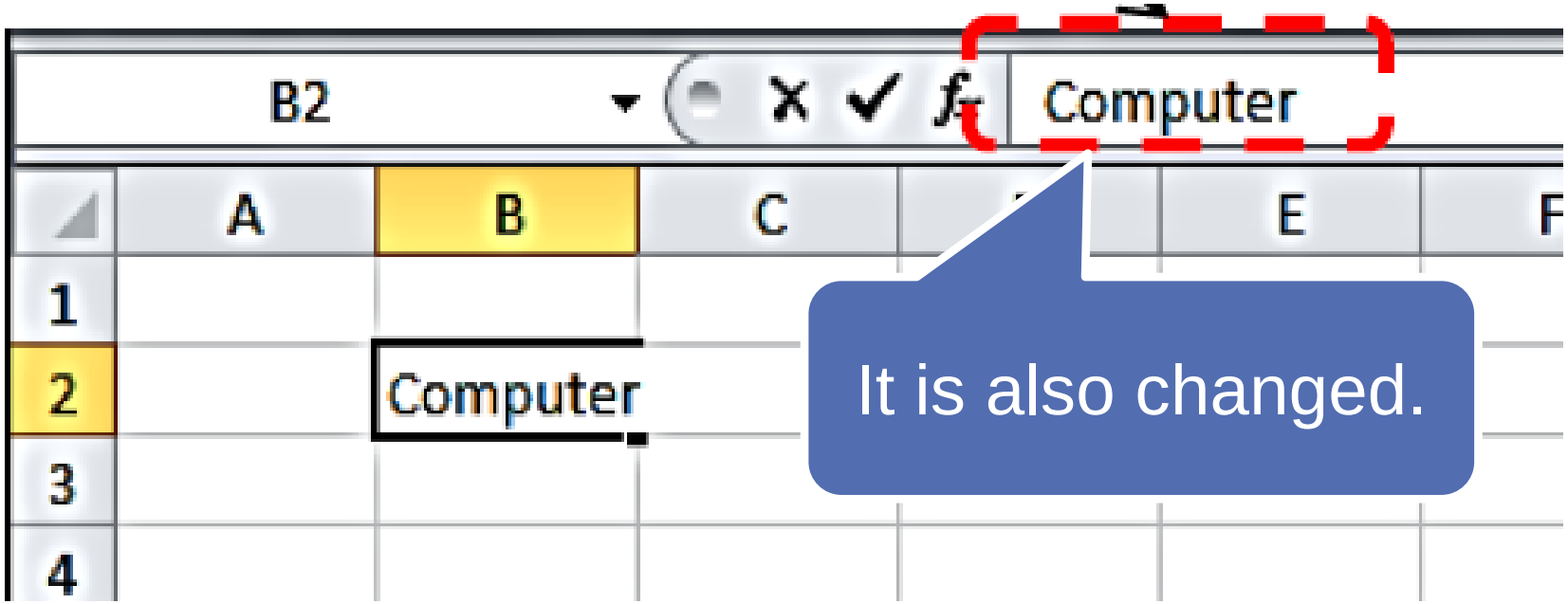
To select a single cell, simply you can ***left click*** the cell. Then a black border surrounds it, and that selected cell becomes the active cell.



ENTERING A DATA IN A CELL

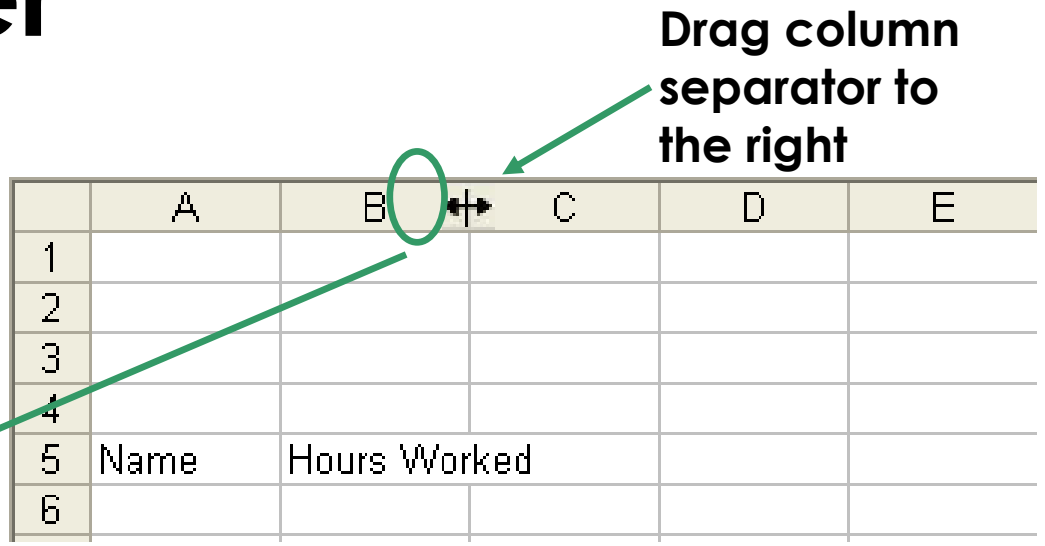
2. Type data

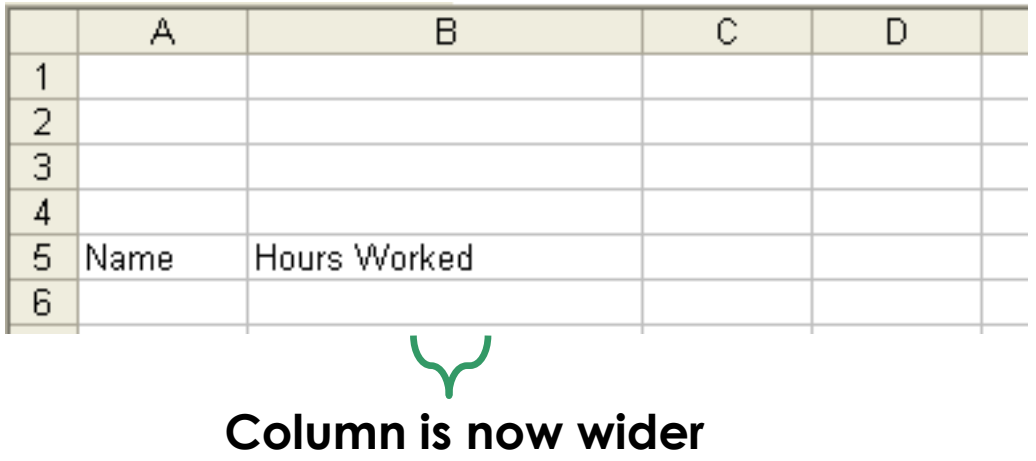
When you enter data, you will find content box is also changed as you typing.



Make a column wider

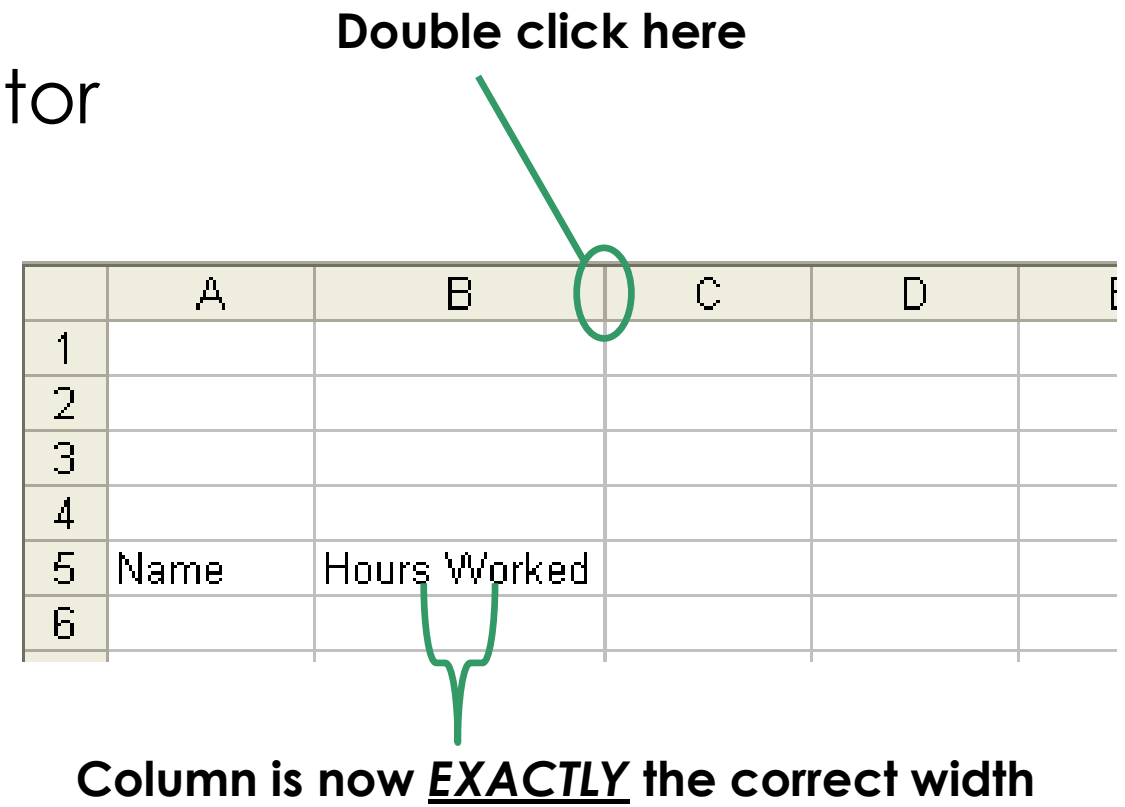
- To make Column B wider, point the cursor to the column separator between columns B and column C.
- The cursor changes to a "Double headed arrow".
- Now, click the left mouse button and without letting go of the button, drag the separator to the right to make the column wider (or to the left to make the column narrower).





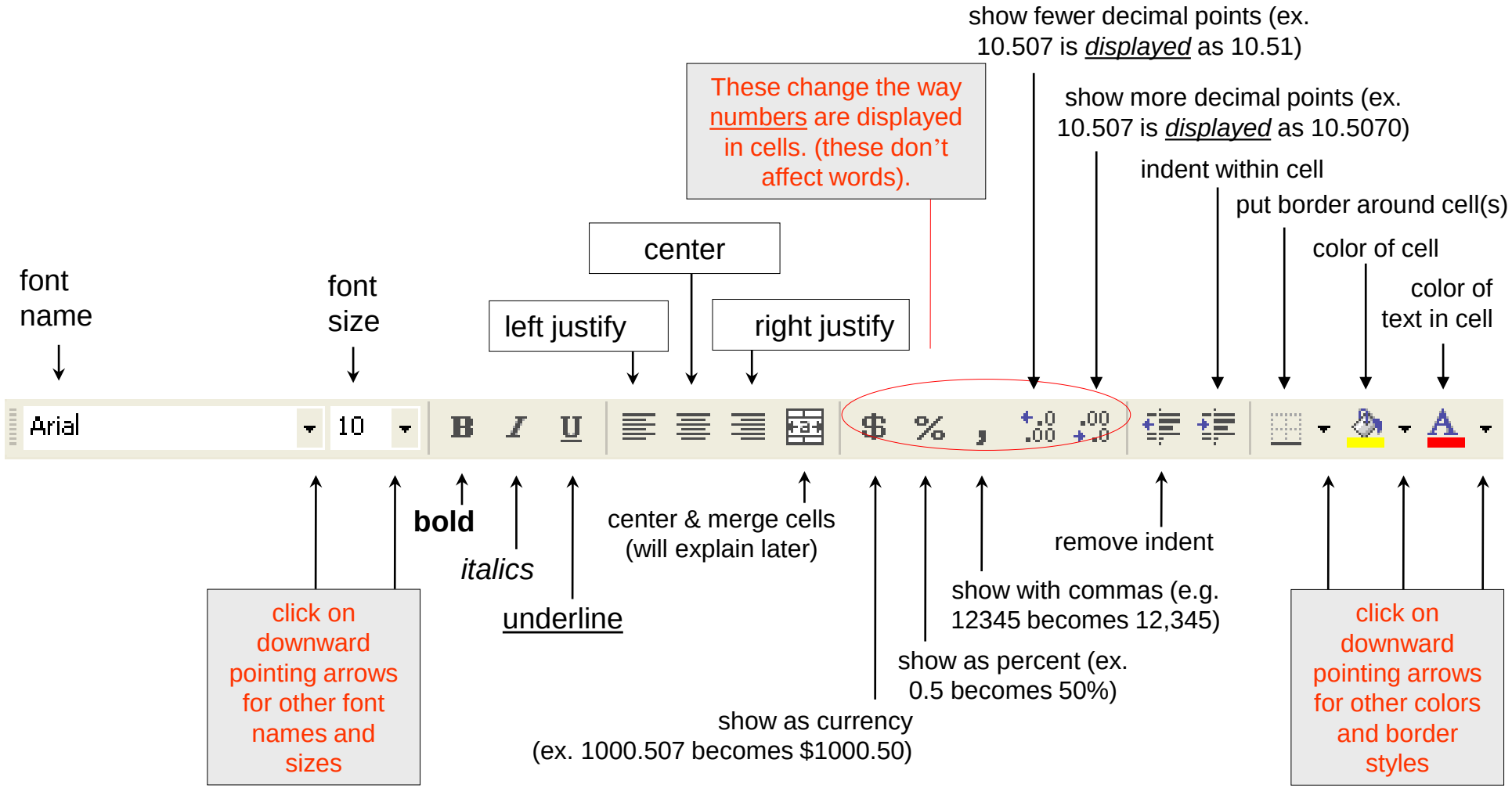
Getting the Exact Width

- To get the "exact" width, double click on the separator instead of dragging it.



Formatting Cells

- Select **one or more cells** and then click on any of the formatting buttons (see below) to change the formatting of the **selected cells**.
- Formatting buttons:



Practical Activity 2

Instructions

1. Open Microsoft Excel
2. Create the table below.

Item	Quantity	Price
Drinks	10	15
Bread	8	12
Biscuits	20	5

Learner Tasks

1. Type text in the correct cells.
2. Enter numbers correctly.
3. Move from one cell to another using: Mouse/Arrow keys/Enter key/Tab key
4. Change “Bread” to “Buns”.
5. Change quantity 20 to 25.
6. Delete the word “Drinks”.
7. Re-enter it correctly.
8. Save the workbook

Borders, colours and text bolding

What is Formatting?

Formatting is the process of changing the appearance of data in a spreadsheet to make it neat, attractive, and easy to read.

In Microsoft Excel, we can format a spreadsheet by:

- Adding **borders**
- Changing **colors**
- Making text **bold**
- Changing **font size** and **style**
- Aligning text



Adding Borders in Excel

What are Borders?

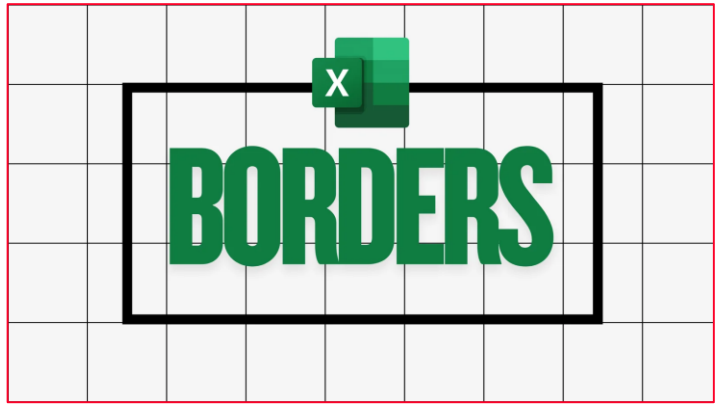
Borders are lines placed around cells to separate and organize data clearly.

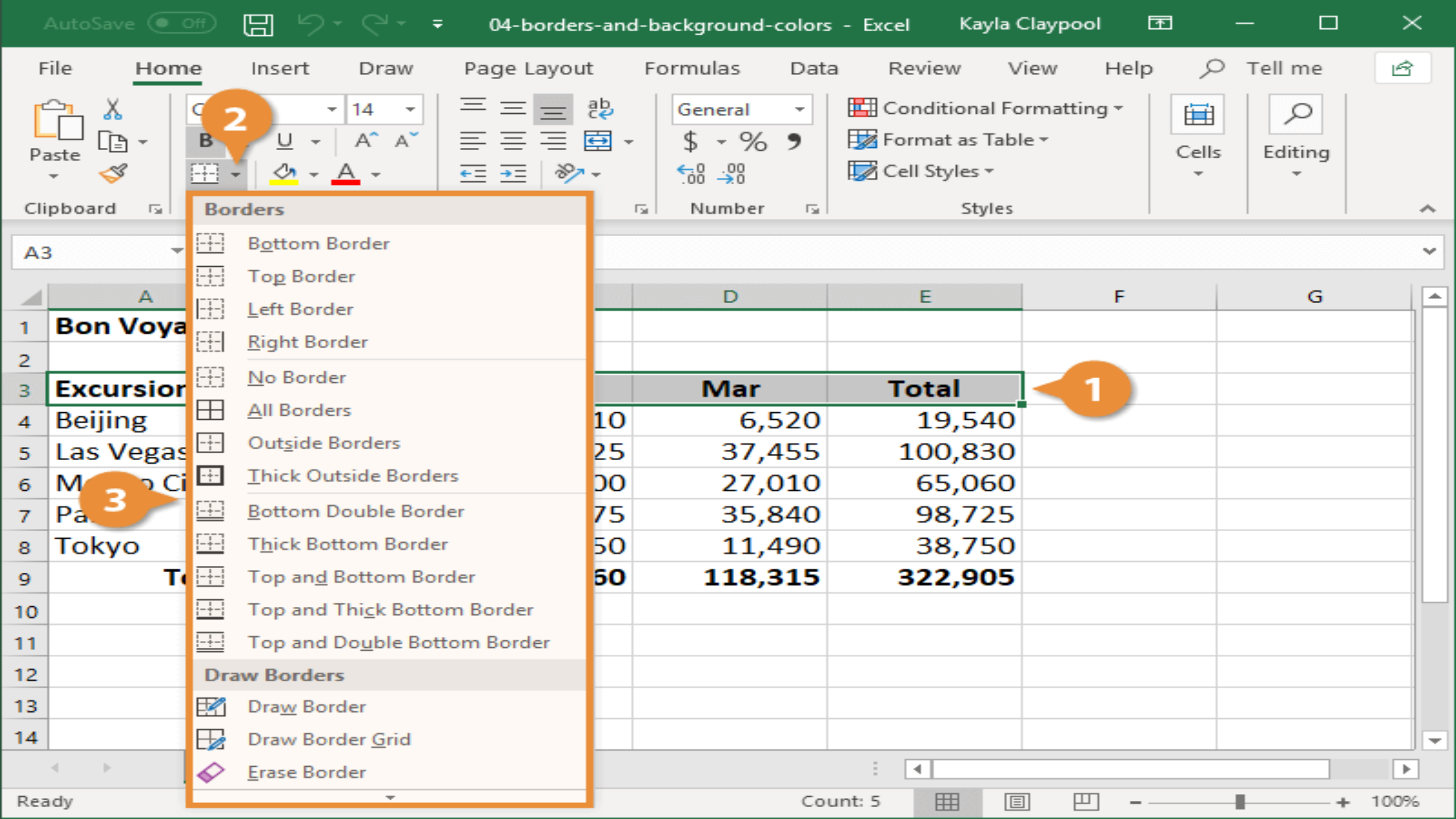
Steps to Add Borders

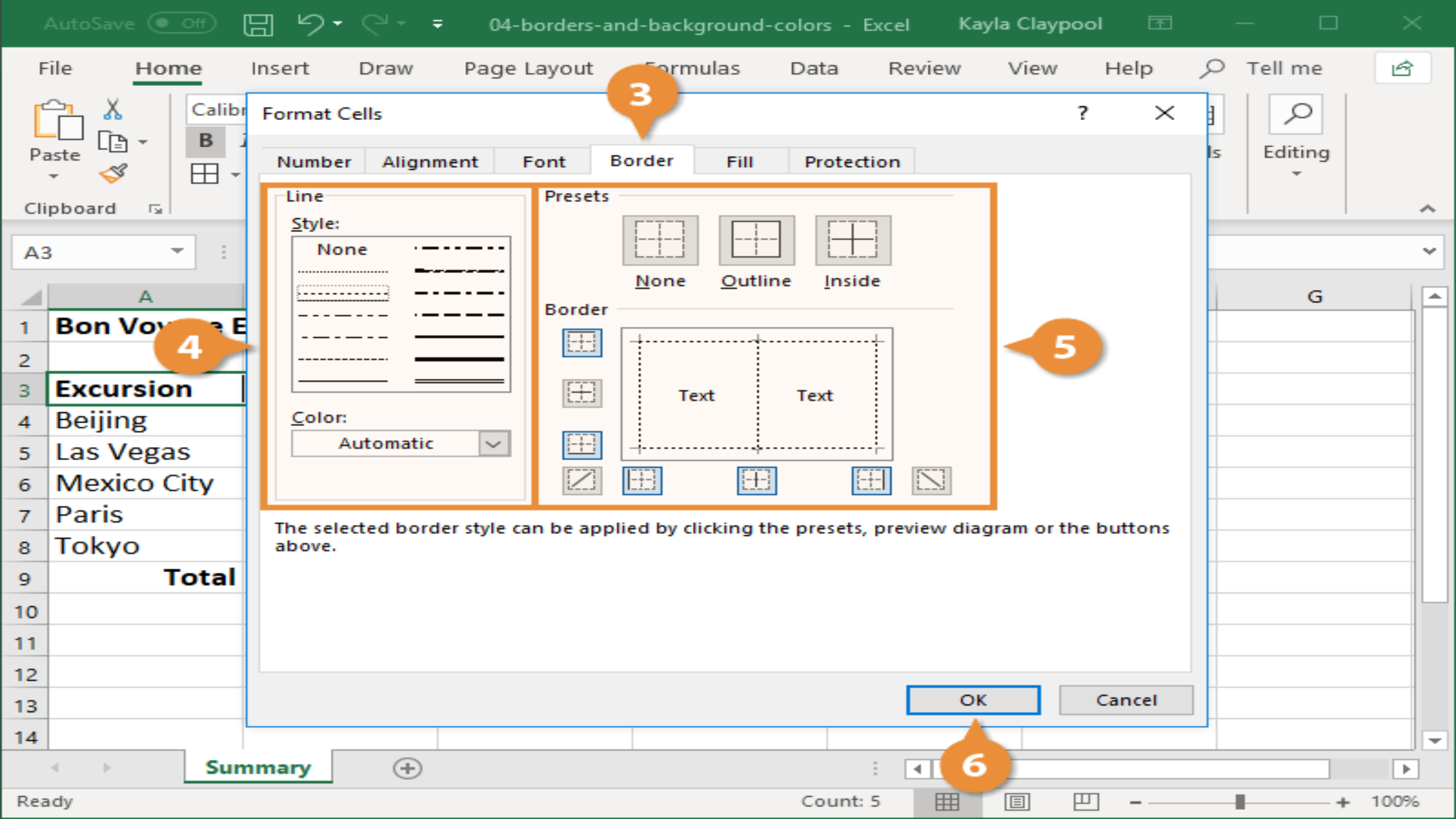
1. Select the cells you want to format.
2. Go to the **Home** tab.
3. Click the **Borders** button in the Font group.
4. Choose a border style such as:
   - All Borders
   - Outside Borders
   - Thick Box Border

Importance of Borders

- Makes tables easy to read
- Separates information clearly
- Improves presentation of work




AutoSave Off
04-borders-and-background-colors - Excel
Kayla Claypool
File Home Insert Draw Page Layout Formulas Data Review View Help Tell me
Paste
Clipboard
14
General
Conditional Formatting
Format as Table
Cell Styles
Cells
Editing
Number
Styles
A3
Borders
Bottom Border
Top Border
Left Border
Right Border
No Border
All Borders
Outside Borders
Thick Outside Borders
Bottom Double Border
Thick Bottom Border
Top and Bottom Border
Top and Thick Bottom Border
Top and Double Bottom Border
Draw Borders
Draw Border
Draw Border Grid
Erase Border
Bon Voya
Excursion
Mar
Total
Beijing 6,520 19,540
Las Vegas 37,455 100,830
27,010 65,060
35,840 98,725
Tokyo 11,490 38,750
118,315 322,905
1
2
3
Ready
Count: 5
100%


AutoSave Off
04-borders-and-background-colors - Excel
Kayla Claypool
File Home Insert Draw Page Layout Formulas Data Review View Help Tell me
Paste
Clipboard
Editing
A3
Format Cells
3
Number Alignment Font Border Fill Protection
Line
Style:
None
Color:
Automatic
4
Presets
None Outline Inside
Border
Text Text
5
The selected border style can be applied by clicking the presets, preview diagram or the buttons above.
OK
Cancel
6
Bon Voyage E
Excursion
Beijing
Las Vegas
Mexico City
Paris
Tokyo
Total
Summary
Ready
Count: 5
100%

Adding Colors in Excel

Types of Colors

- **Fill Color -** changes the background color of a cell
- **Font Color** - changes the text color

Steps to Add Fill Color

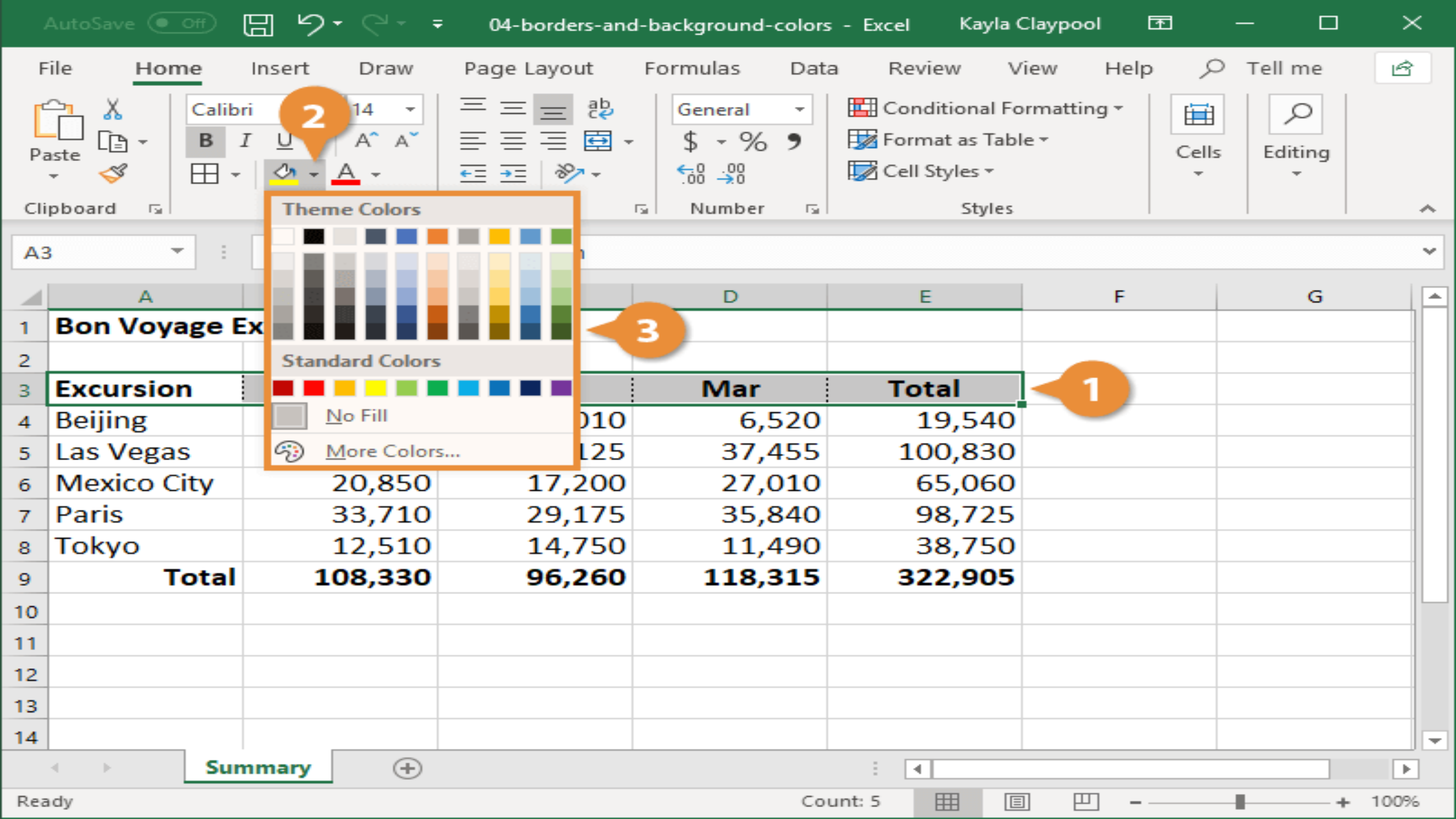
1. Select the cells.
2. Click Fill Color on the Home tab.
3. Choose the desired color.

Steps to Change Font Color

1. Select the text or cells.
2. Click Font Color.
3. Choose a color.

Importance of Colors

- Highlights important information
- Makes the spreadsheet attractive
- Helps users identify sections quickly


AutoSave Off
04-borders-and-background-colors - Excel
Kayla Claypool
File Home Insert Draw Page Layout Formulas Data Review View Help Tell me
Paste
Calibri
14
B I U
General
Conditional Formatting
Format as Table
Cell Styles
Cells
Editing
Clipboard
Number
Styles
2
Theme Colors
Standard Colors
No Fill
More Colors...
3
1
A3
Bon Voyage Ex
Excursion
Mar
Total
Beijing 6,520 19,540
Las Vegas 37,455 100,830
Mexico City 20,850 17,200 27,010 65,060
Paris 33,710 29,175 35,840 98,725
Tokyo 12,510 14,750 11,490 38,750
Total 108,330 96,260 118,315 322,905
Summary
Ready
Count: 5
100%

Making Text Bold

What is Bold Text?

Bold text is darker and thicker than normal text.

Steps to Make Text Bold

1. Select the text or cells.
2. Click the Bold (B) button on the Home tab. or
3. Press Ctrl + B on the keyboard.

Importance of Bold Text

- Emphasizes headings
- Makes important data stand out
- Improves readability

Activity 3: Practical Task

Instructions

Open Microsoft Excel and create the table below.

	A	B	C	D
1	**Subject**	**Boys**	**Girls**	**Tuition Fees**
2	ICT	12	15	All Paid
3	English	10	14	14 paid and were all girls
4	Science	11	13	All paid

Task

1. Add borders to all cells.
2. Make the heading row bold.
3. Add a color to the heading row.
4. Change the font color of the headings.
5. Save the work as Class Payments.

Group Work

In groups, discuss:

- Why formatting is important in spreadsheets.
- Situations where borders, colors, and bold text are useful.



Activity 4 - Scenario 1

The head teacher at a school asked Form 2 learners to prepare a spreadsheet showing sports club members. The spreadsheet looked confusing because there were no borders, headings were not bold, and all cells had the same color.

Questions

1. Which formatting tools can learners use to improve the spreadsheet?
2. Why is bold text important in headings?
3. How do borders help in a spreadsheet?
4. Why should colors be used carefully?

Activity 5 - Scenario 2

The Grade 12 class at your school was asked to prepare a spreadsheet showing money collected for a school trip.

After entering the data, Mr. Hakubeza noticed that the spreadsheet was difficult to read because the headings were not bold, rows and columns were not separated, and important totals were not highlighted.

Mr. Hakubeza instructed the learners to improve the spreadsheet by:

- Applying bold text to headings and key totals
- Inserting borders to clearly separate rows and columns
- Using colours to highlight important sections such as totals, categories, and low funds
- Ensuring the spreadsheet is neat, readable, and logically arranged

Spreadsheet Example

Category	Amount Collected	Amount Needed	Balance
Transport	K800	K1000	K200
Food	K600	K700	K100
Uniforms	K500	K900	K400
Total	K1900	K2600	K700

Tasks

1. Create the table in Microsoft Excel.
2. Make all headings bold and center align
3. Apply bold formatting to the Total row.
4. Insert borders around all cells.
5. Use different colours to:
   - Highlight headings
   - Highlight the Total row
   - Show low balances or low funds
6. Arrange the spreadsheet neatly for easy reading.
7. Save the file as **School Trip Budget.**

Questions

1. Why should headings be bold in a spreadsheet?
2. How do borders improve the appearance of a spreadsheet?
3. Why is it important to use colours in spreadsheets?
4. What could happen if a spreadsheet is not arranged neatly?
5. Which section would you highlight to show low funds?

Sorting Data in Spreadsheets

Meaning of Sorting

Sorting means arranging data in a specific order.

Types of Sorting

1. Ascending Order
2. Descending Order

- **Ascending Order**
  - A to Z
  - Smallest to largest
  - Example: 10, 20, 30
- **Descending Order**
  - Z to A
  - Largest to smallest
  - Example: 90, 70, 50

Steps for Sorting Data

- Highlight the table/data.
- Click the **Data** tab.
- Choose:
  - **Sort A–Z** for ascending order
  - **Sort Z–A** for descending order

Importance of Sorting Data

Sorting helps users:

- Find information quickly
- Arrange records neatly
- Compare data easily
- Improve organization of work

Example

- **Before Sorting**

Name	Marks
Jane	78
Brian	65
Bupe	82

- **Example After Sorting (Highest to Lowest)**

Name	Marks
Bupe	82
Jane	78
Brian	65

Activity 6: Sorting Data

Task 1: Entering Data

Create a spreadsheet showing:

- Province
- Town
- Population

- Enter data for 10 Provinces.

Task 2: Sorting Data

Using the population entered:

1. Sort the numbers from highest to lowest.
2. Sort the town names alphabetically.

Scenario-Based Activity

Your teacher of Science recorded test marks using a spreadsheet. Some learners wanted to know who scored the highest and the lowest marks.

Questions

1. What feature can the teacher use to arrange the marks?
2. What is ascending order?
3. What is descending order?
4. Why is sorting important in spreadsheets?

Using the MIN, MAX, COUNT, COUNTA, SUM-IF functions

In spreadsheets, functions are special formulas used to perform calculations quickly and accurately. Some commonly used functions in Microsoft Excel are **MIN, MAX, COUNT, COUNTA, and SUMIF**. The **MIN** function finds the smallest value, while **MAX** finds the largest value in a range of numbers. **COUNT** counts cells containing numbers only, whereas **COUNTA** counts all non-empty cells including text and numbers. The **SUMIF** function adds values that meet a specific condition.

These functions help learners organize, analyze, and manage data easily in everyday tasks such as budgeting, recording marks, and keeping school records.

Calculations in Excel

Formulas

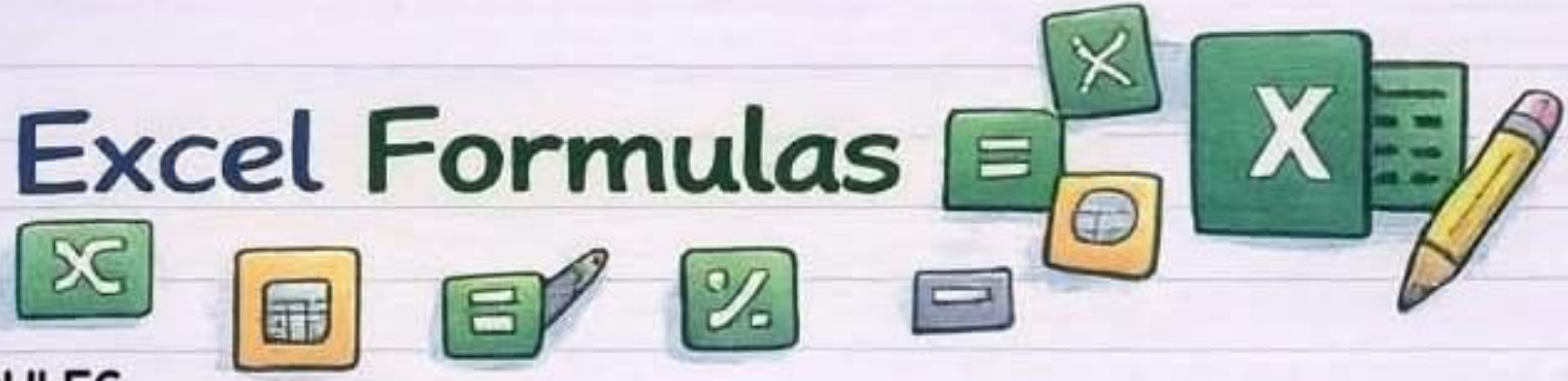
In Excel, a formula is an expression that operates on values in a range of cells or a cell. For example, =A1+A2+A3, which finds the sum of the range of values from cell A1 to Cell A3.

Functions

Functions are predefined formulas in Excel. They eliminate laborious manual entry of formulas while giving them human-friendly names. For example: =SUM(A1:A3). The function sums all the values from A1 to A3.

NOTE: *Always starts a formula by typing an equal sign =.*
= sign that tells Excel that what is in the cell is a formula and not a text or a number.

Excel Formulas



RULES

- Everg formula must start with an equals sign (=).
  Eg. =A1+B1
- Use correct cell references, using cell name (e.g., A1, B2:B10).
- Follow the order of operations (BODMA5)
- Use commas or semicolons correctly e g, =SUM(A1,A2)
- Text values must be in quotes. Examole: =IF(A1='Yes",
- Avoid unnecessary spaces inside formulas

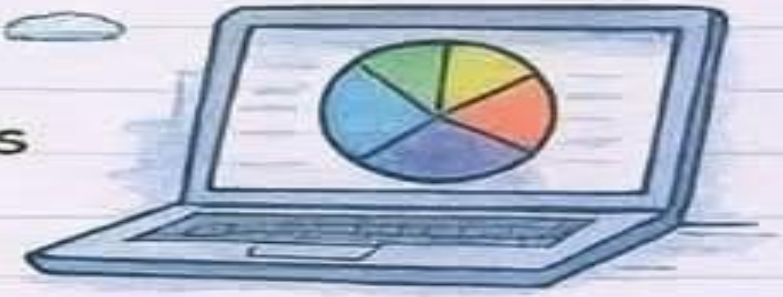
Common Excel Formulas

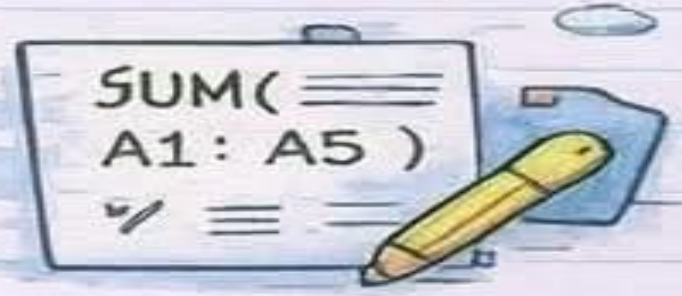
- SUM – Adds values together
  =SUM(A1:A5)
- AVERAGE – Finds the mean value
  =AVERAGE(A1:A5)
- COUNT – Counts cells with numbers
  =COUNT(A1:A5)
- COUNTA – Counts non-empty cells
  =COUNTA(A1:A5)
- MIN – Returns the smallest value
  =MIN (A1:A5)
- MAX – Returns the largest value
  =MAX (A1:A5)



Formulas





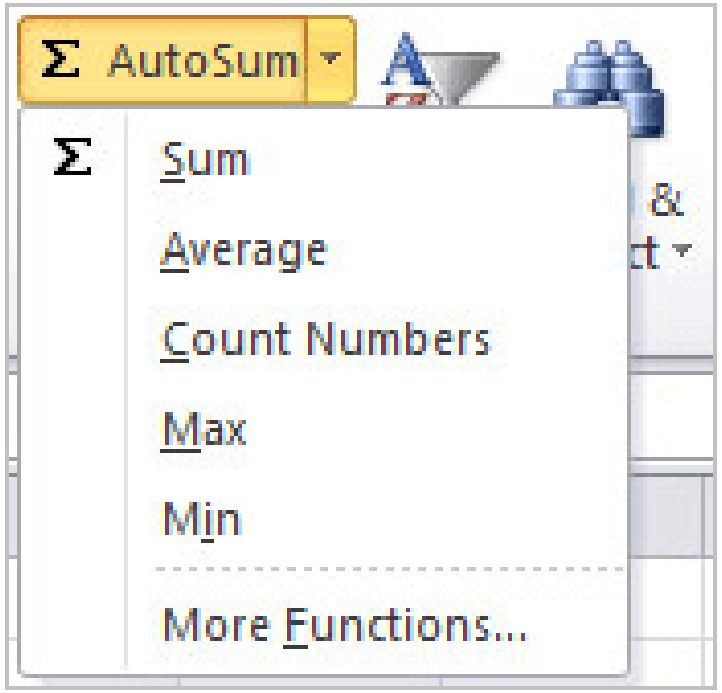




Importance of these Functions

These functions help learners to:

- Analyze data easily
- Save time when calculating
- Reduce calculation errors
- Manage records efficiently in spreadsheets



SUM Function

Meaning

The **SUM** function is used to add values together.

Example

=SUM(B2:B6)

Use

Used to:

- Calculate total marks
- Add expenses in a budget
- Find total sales

	A	B	C	D
1		Jan	Feb	
2	**Entertainment**			
3	Cable TV	52.98	52.98	
4	Video Rentals	7.98	11.97	
5	Movies	16.00	32.00	
6	CDs	18.99	29.99	
7	Totals	=SUM(B3:B6)		
8				

Scenario: Using the SUM Function

Thabo just arrived at a boarding school and he recorded the money spent during a shopping day in a spreadsheet.

He wants to calculate the total amount spent using the SUM function.

Item	Amount (K)
Transport	400
Food	710
Shoes	375
Uniform	900

Task

1. Enter the data into a spreadsheet.
2. Use the SUM function to calculate the total expenses.

Formula Example

=SUM(B2:B5)

Questions

a. What is the purpose of the SUM function?
b. Which cells contain the amounts?
c. What is the total amount spent?

AVERAGE Function

Meaning

The **AVERAGE** function calculates the mean of selected numbers.

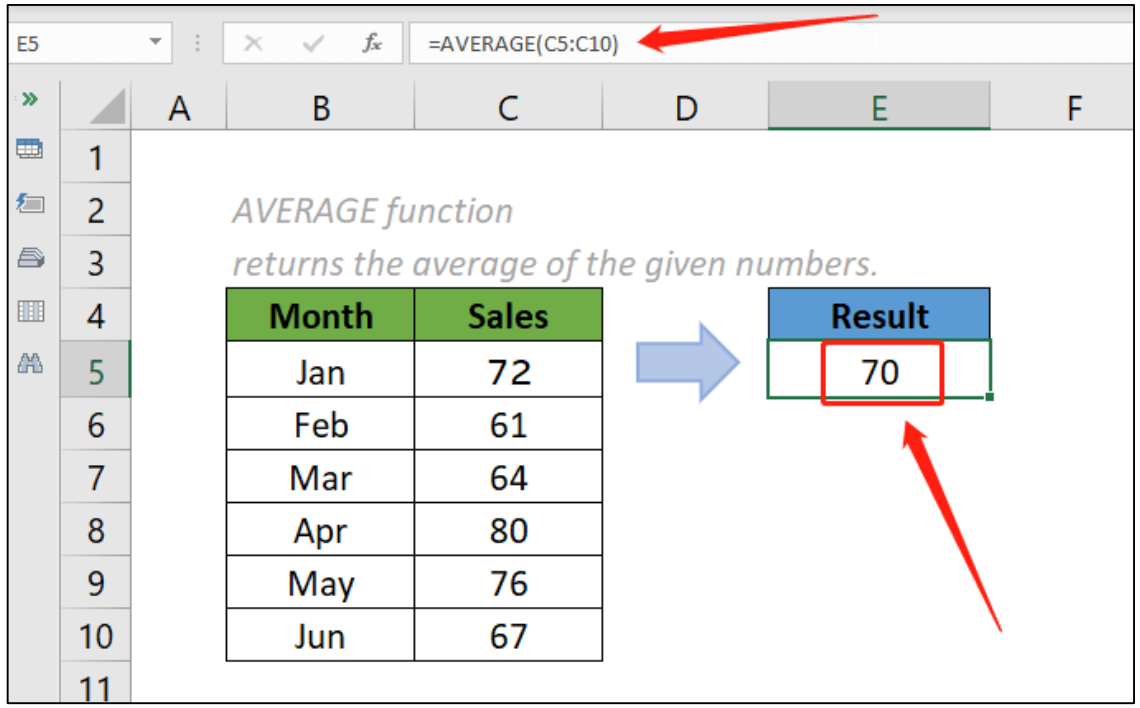
Example

=AVERAGE(B2:B6)

Use

Used to:

- Calculate average marks
- Find average expenses
- Analyze performance



MIN Function

Meaning

The **MIN** function finds the **smallest value** in a group of numbers.

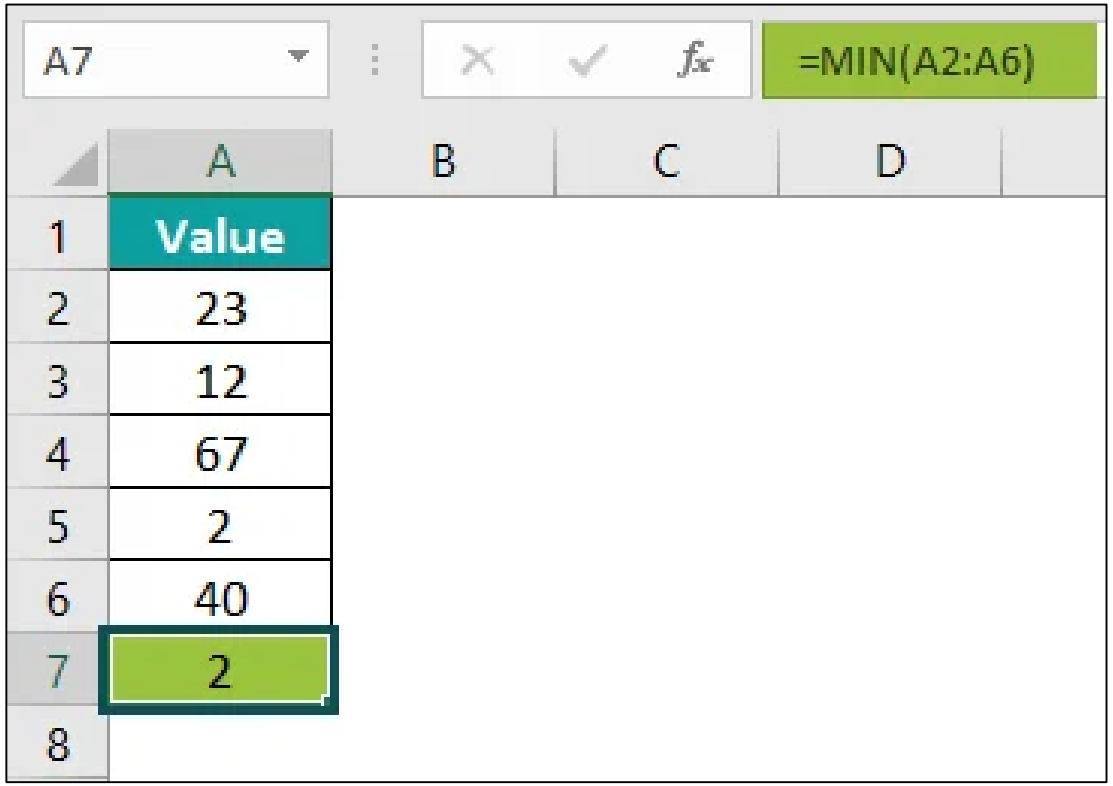
Example

=MIN(B2:B6)

Use

Used to find:

- Lowest mark
- Smallest expense
- Lowest price



Scenario: Using the MIN Function

A teacher entered the marks of five learners into a spreadsheet as shown below:

Leaner	Marks
Grace	78
John	65
Twalumba	82
Karen	54
Chipo	70

The teacher wants to find the lowest mark in the class using a spreadsheet function.

Task

1. Which spreadsheet function should be used to find the lowest mark?
2. Write the formula that can be used.
3. What is the lowest mark?

Formula

=MIN(B2:B6)

MAX Function

Meaning

The **MAX** function finds the **largest value** in a group of numbers.

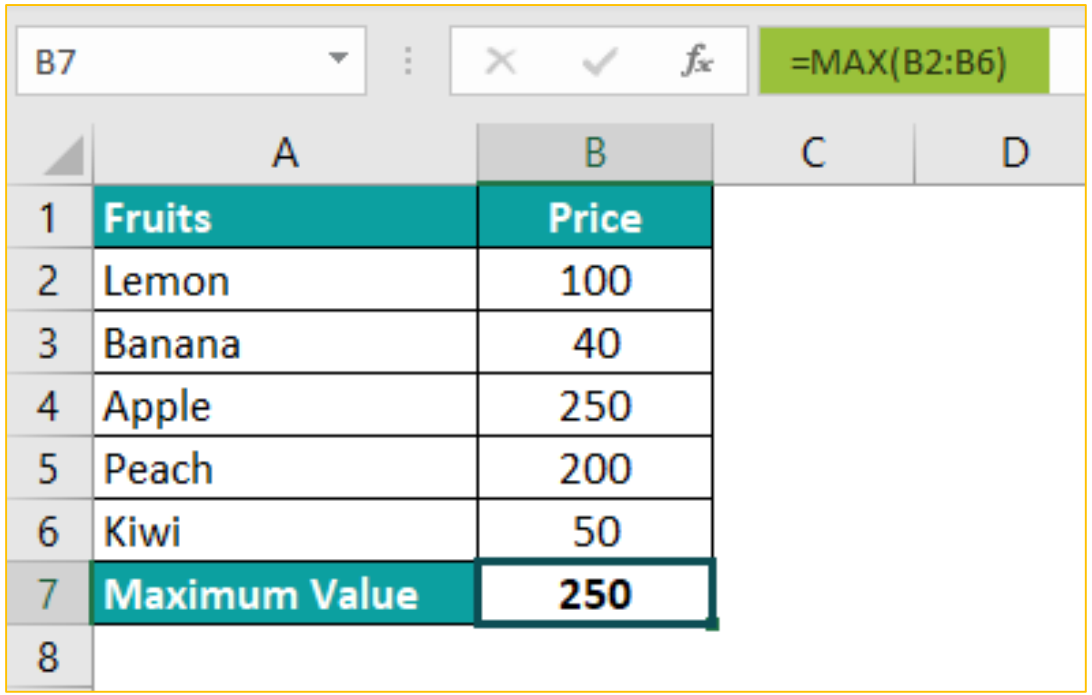
Example

=MAX(B2:B6)

Use

Used to find:

- Highest mark
- Largest amount
- Highest score



COUNT Function

Meaning

The **COUNT** function counts cells containing numbers only.

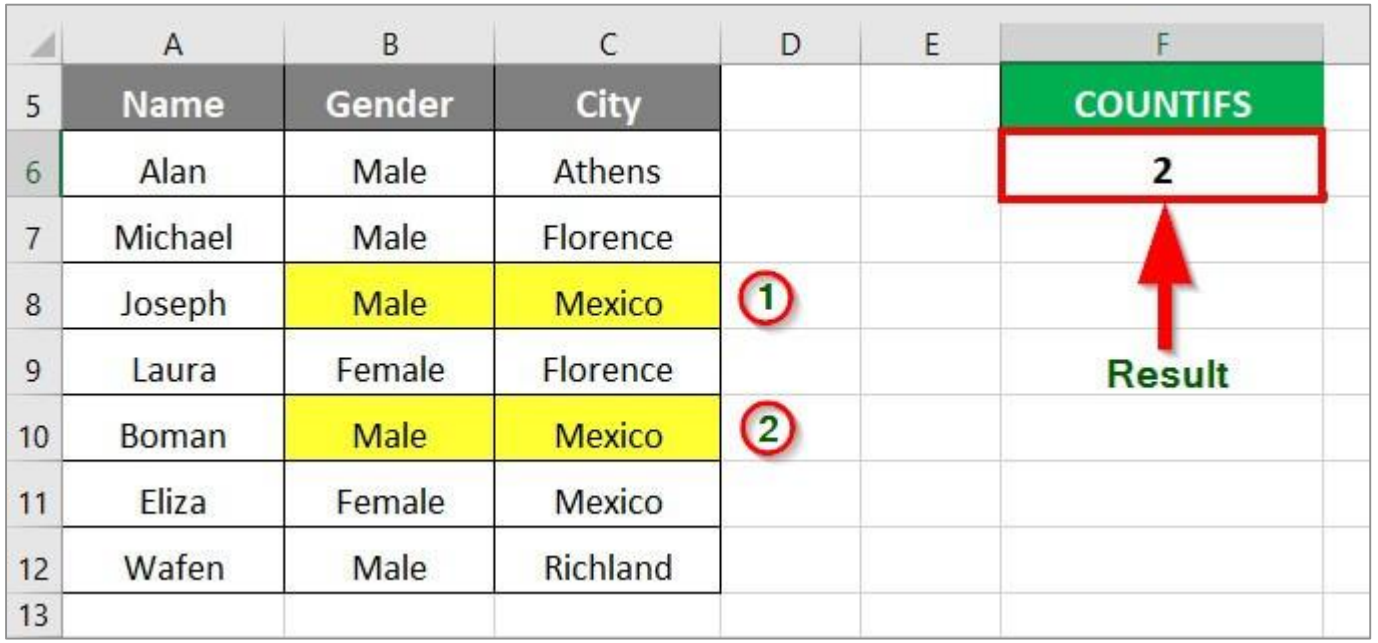
Example

=COUNT(B2:B10)

Use

Used to count:

- Number of learners with marks
- Numeric entries in a spreadsheet



	A	B	C	D	E	F
5	Name	Gender	City			COUNTIFS
6	Alan	Male	Athens			2
7	Michael	Male	Florence			
8	Joseph	Male	Mexico	1		
9	Laura	Female	Florence			Result
10	Boman	Male	Mexico	2		
11	Eliza	Female	Mexico			
12	Wafen	Male	Richland			
13						

COUNTA Function

Meaning

The **COUNTA** function counts all non-empty cells:

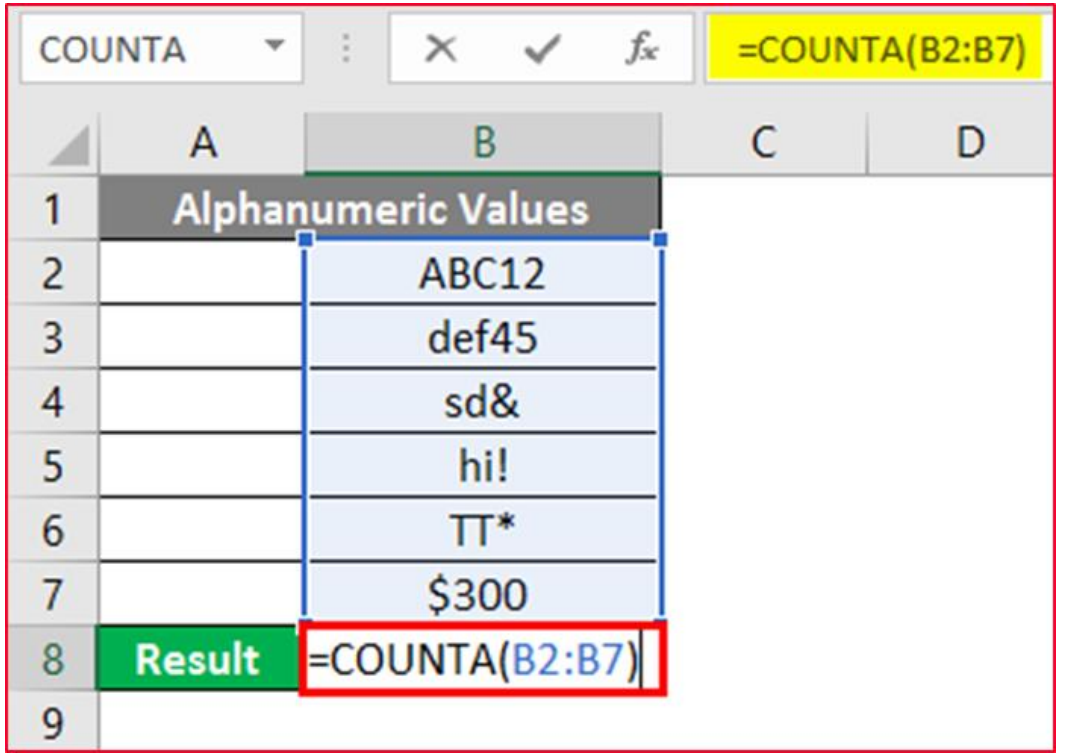
- Numbers
- Words
- Dates

Example

=COUNTA(B2:B7)

Use

- Used to count:
- All entered records
- Names or text entries



SUM-IF Function

Meaning

The **SUMIF** function adds numbers that meet a specific condition.

Example

=SUMIF(A2:A10,"Food",B2:B10)

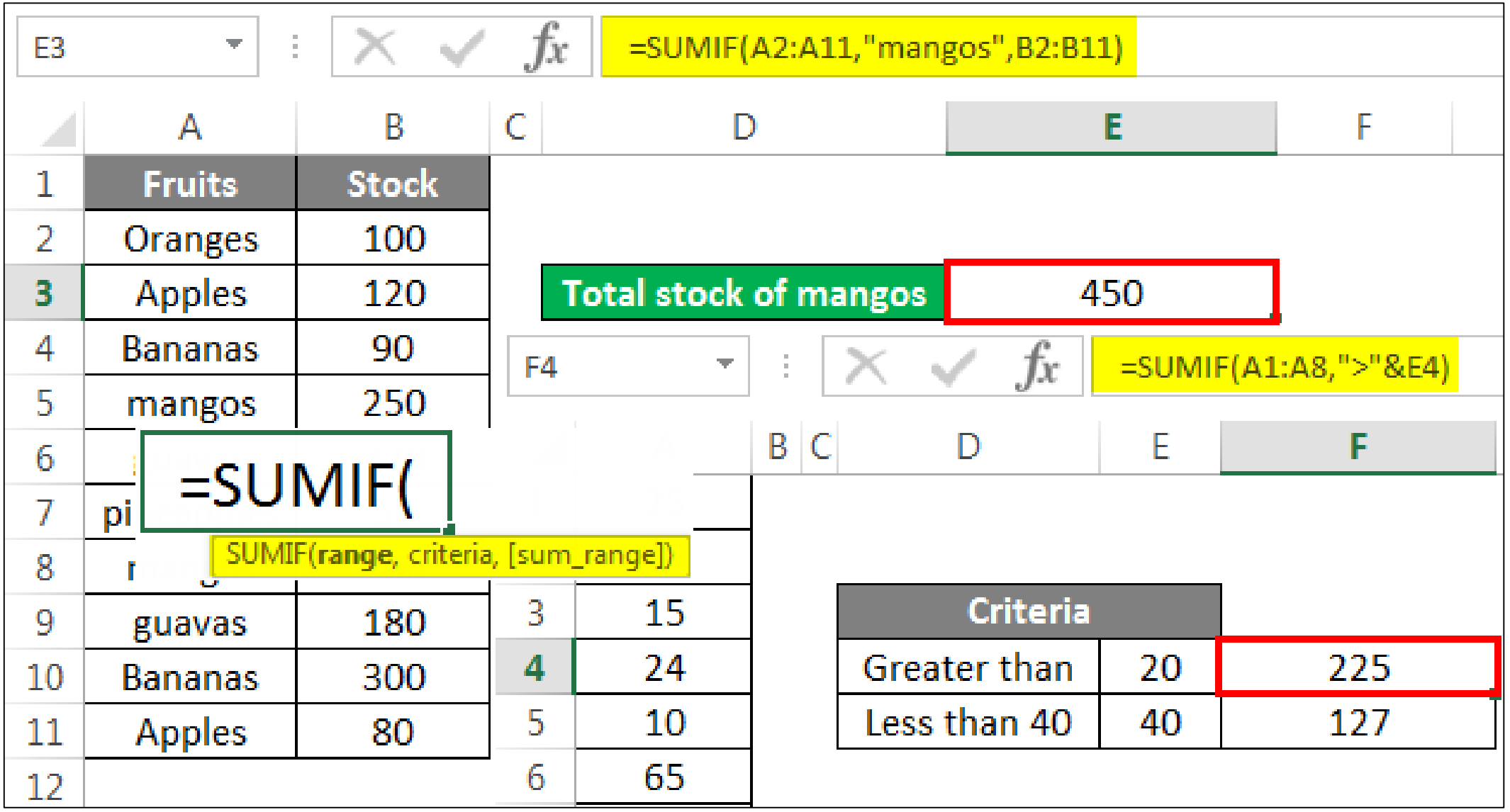
Meaning of the Formula

- Check cells A2 to A10
- Find the word “Food”
- Add corresponding amounts from B2 to B10

Use

Used to:

- Total transport expenses only
- Add marks for one subject
- Calculate category totals


E3
=SUMIF(A2:A11,"mangos",B2:B11)
A
B
C
D
E
F
1 Fruits Stock
2 Oranges 100
3 Apples 120
4 Bananas 90
5 mangos 250
6
7 pi
8 r
9 guavas 180
10 Bananas 300
11 Apples 80
12
Total stock of mangos
450
F4
=SUMIF(A1:A8,">"&E4)
=SUMIF(
SUMIF(range, criteria, [sum_range])
3 15
4 24
5 10
6 65
Criteria
Greater than 20 225
Less than 40 40 127

Creating a budget using spreadsheet

A budget is a plan that shows how money will be earned and spent over a certain period of time. A spreadsheet helps us organize calculations neatly and automatically.

Learners can use spreadsheets to:

- Plan spending
- Calculate totals automatically
- Compare income and expenses
- Save time and reduce mistakes



Key Terms

Term	Meaning
Budget	A plan for using money
Income	Money received
Expense	Money spent
Spreadsheet	A program used to organize and calculate data
Formula	An equation used for calculations

Example: Monthly Student Budget

A	B
Item	**Amount (K)**
Pocket Money	300
Lunch	120
Transport	80
Airtime	40
Savings	60

Steps for Creating a Budget in Excel

Step 1: Open the Spreadsheet Program

Open:

- Microsoft Excel or
- Google Sheets

Step 2: Enter Headings

Type:

- Item
- Amount

Step 3: Enter Budget Data

Fill in income and expenses in rows below the headings.

Step 4: Use a Formula to Calculate Total Expenses

- Click in the cell below the amounts and type:

=SUM(B2:B5)

- Press **Enter**.
- The spreadsheet automatically calculates the total.

Classroom Activity:
Create your own weekly budget

Imagine you receive K750 for one week.

Create a spreadsheet showing:

- Income
- Transport
- Snacks
- Airtime
- Savings

Then:

1. Calculate total expenses.
2. Find how much money remains.
3. Format the spreadsheet neatly.

Scenario-Based Task

Chuma is preparing for a school trip. Her parents gave her K500. She plans to spend money on:

- Food
- Transport
- Drinks
- Emergency savings

Create a spreadsheet budget showing:

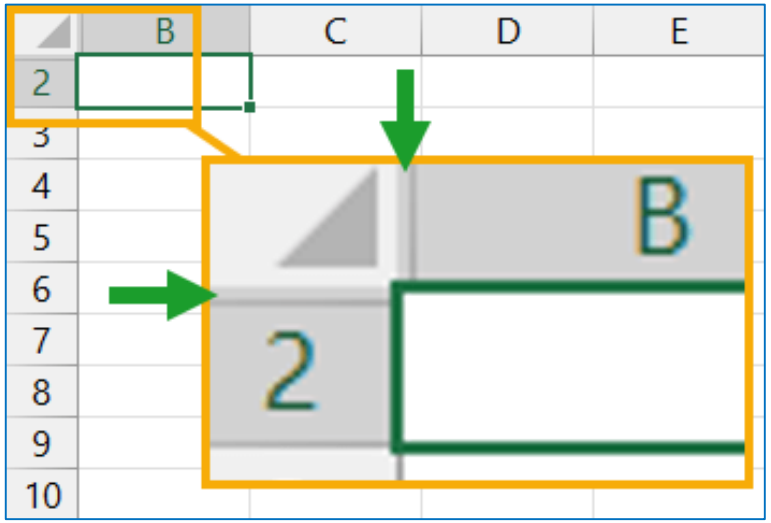
- The items
- The amount for each item
- Total money spent
- Remaining balance

Assessment Questions

1. What is a budget?
2. Name two uses of spreadsheets in budgeting.
3. Which formula is used to calculate totals?

Hiding and Un-hiding rows and columns

- Sometimes a spreadsheet contains information that we do not want to see all the time.
- A spreadsheet allows users to **hide** rows and columns without deleting the data. The hidden information can later be **un-hidden** when needed.
- This helps keep the worksheet neat and organized.



Key Terms

Term	Meaning
Row	Horizontal arrangement of cells identified by numbers
Column	Vertical arrangement of cells identified by letters
Hide	Temporarily remove data from view without deleting it
Un-hide	Make hidden rows or columns visible again

Importance of Hiding Rows and Columns

Hiding rows and columns helps users to:

- Organize information neatly
- Focus on important data
- Protect unnecessary details from view
- Make worksheets easier to read

Hiding and un-hiding rows and columns are useful spreadsheet skills that help users organize and manage information effectively without deleting data.

Hiding and Un-Hiding Rows

Hiding Rows

Steps for Hiding a Row

1. Select the row number.
2. Right-click the selected row.
3. Click **Hide**.

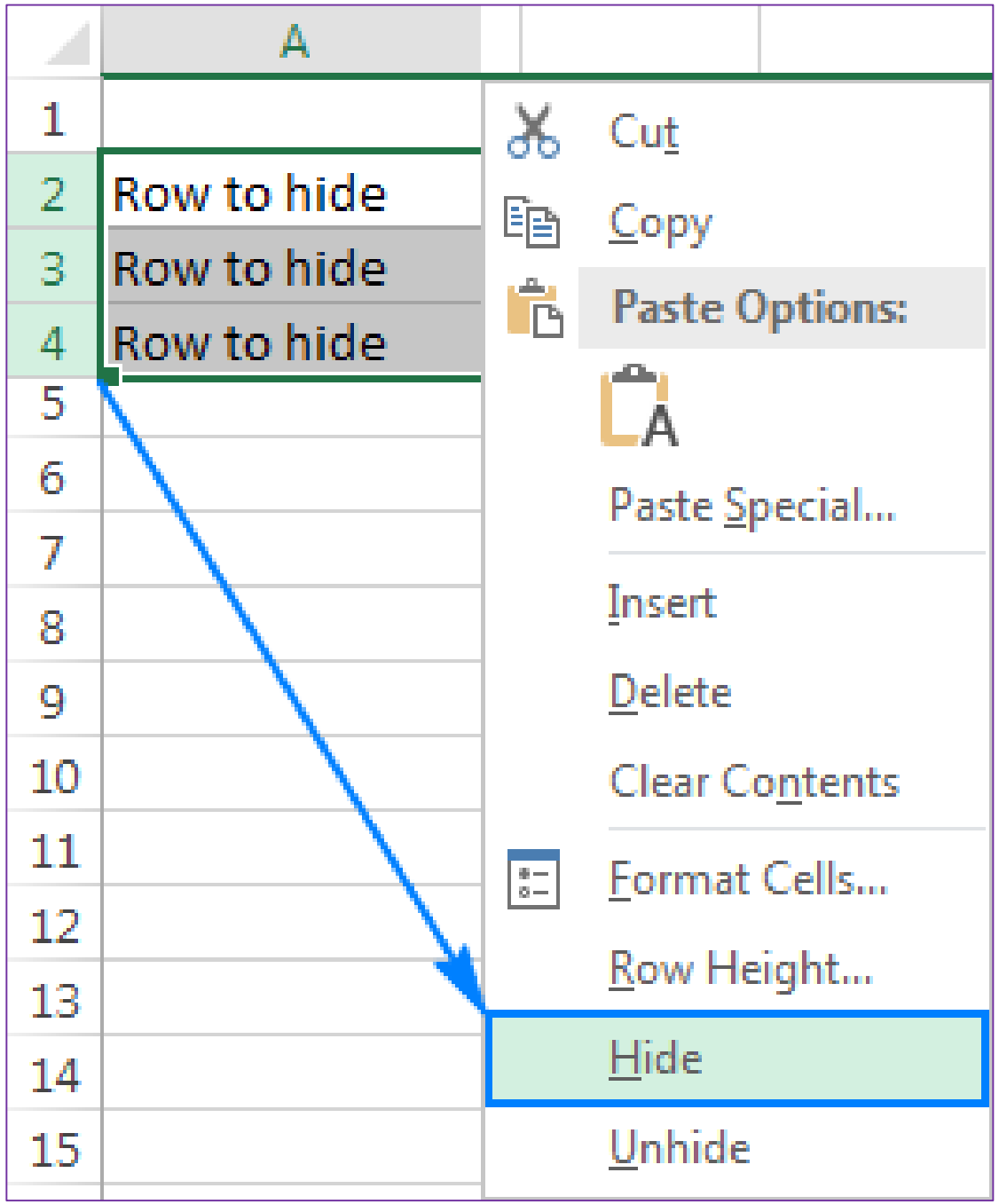
- The row disappears from the worksheet but the data is not deleted.

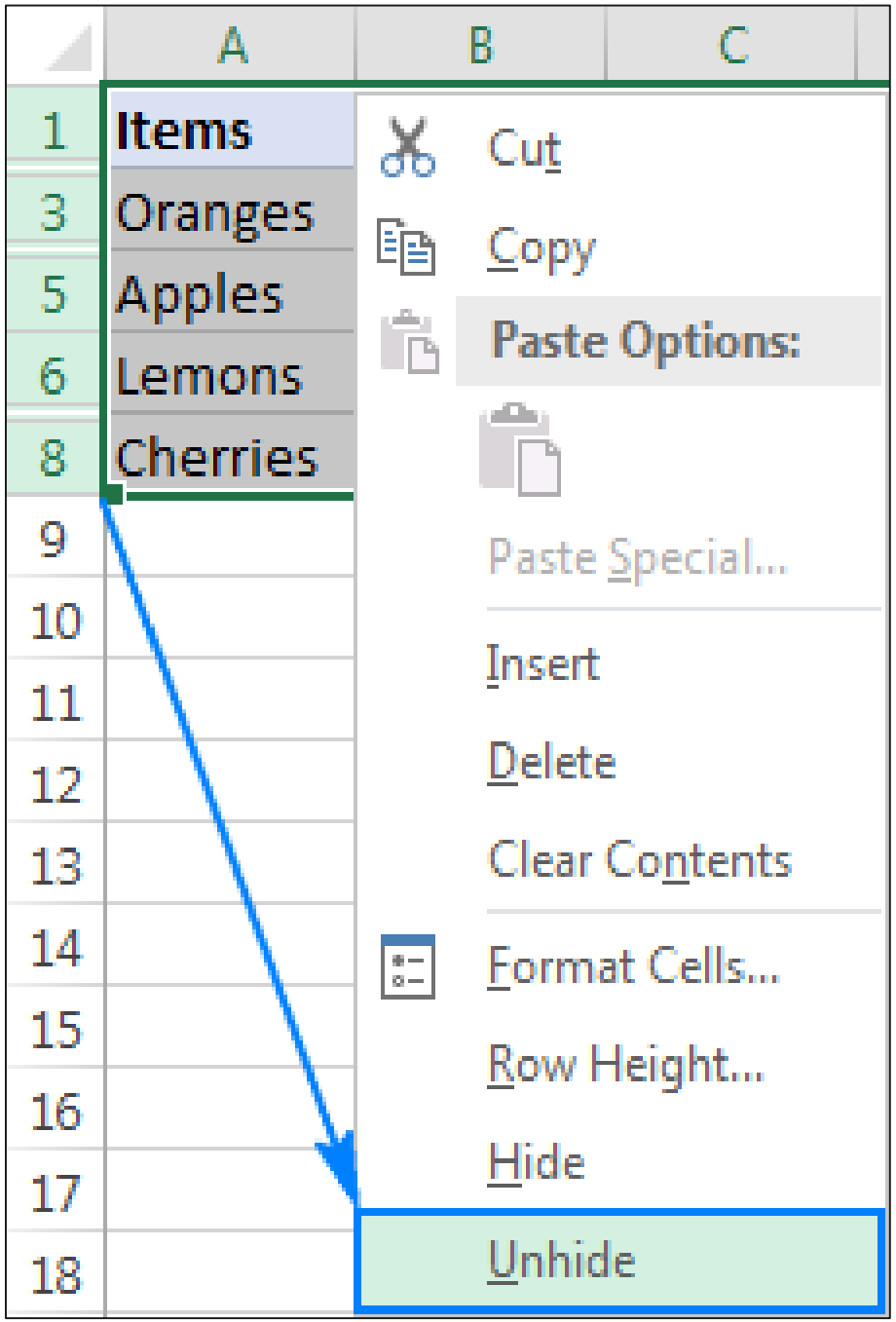
Un-Hiding Rows

Steps for Un-Hiding Rows

1. Select the rows above and below the hidden row.
2. Right-click the selected area.
3. Click **Unhide**.

- The hidden row becomes visible again.


A
1
2 Row to hide
3 Row to hide
4 Row to hide
5
6
7
8
9
10
11
12
13
14
15
Cut
Copy
Paste Options:
Paste Special...
Insert
Delete
Clear Contents
Format Cells...
Row Height...
Hide
Unhide


A
B
C
1 Items
3 Oranges
5 Apples
6 Lemons
8 Cherries
9
10
11
12
13
14
15
16
17
18
Cut
Copy
Paste Options:
Paste Special...
Insert
Delete
Clear Contents
Format Cells...
Row Height...
Hide
Unhide

Scenario-Based Task: Hiding and Un-hiding Rows

A Clerk at a Local Court has created a spreadsheet showing **cases' of appeal the first month of the year 2026**. The spreadsheet includes:

- Complainants names
- Defendant names
- Witness names
- Judge remarks

During a court hearing session, the Judge wants audience to **focus only on the witnesses and complainants narration**.

The "Remarks" section contains private feedback that should not be shown at that moment.

However, later in the session, the Judge needs to show the remarks again when passing on the judgement.

Task

1. Identify the row(s) that should be hidden if the "Remarks" are in row 6 of the spreadsheet.
2. Describe the steps the judge should follow to hide row 6.
3. Explain how the judge can un-hide row 6 after the discussion.
4. State one reason why hiding rows is useful in this situation.
5. If the judge accidentally hides the wrong row (for example row 4 instead of row 6), what should be done to correct the mistake?

Hiding and Un-Hiding Columns

Hiding Columns

Steps for Hiding a Column

1. Select the column letter.
2. Right-click the selected column.
3. Click **Hide**.

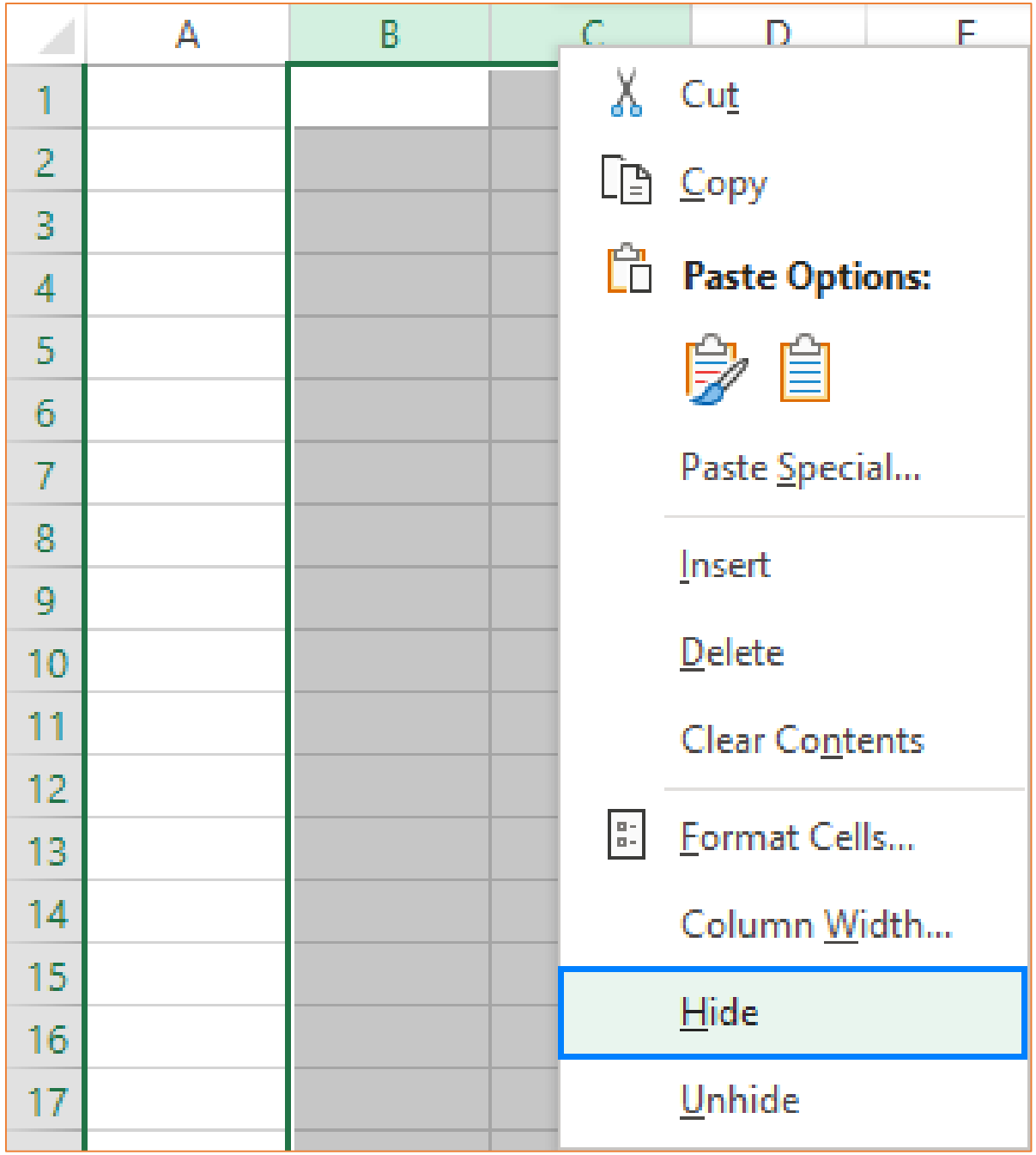
- The column disappears from view.

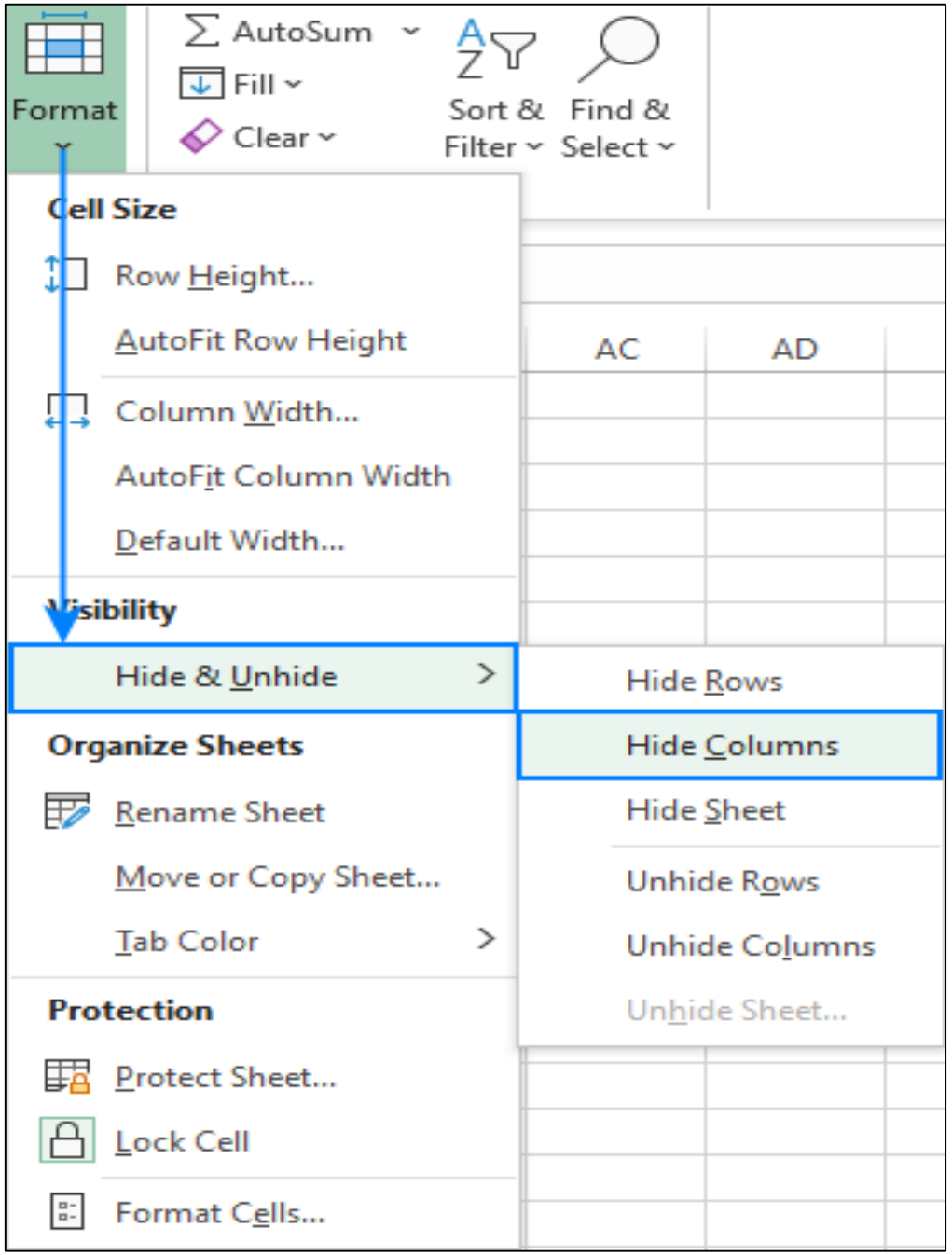
Un-Hiding Columns

Steps for Un-Hiding Columns

1. Select the columns on both sides of the hidden column.
2. Right-click the selected area.
3. Click **Unhide**.

- The hidden column appears again.


A
B
C
D
F
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
Cut
Copy
Paste Options:
Paste Special...
Insert
Delete
Clear Contents
Format Cells...
Column Width...
Hide
Unhide


Format
AutoSum
Fill
Clear
Sort & Filter
Find & Select
Cell Size
Row Height...
AutoFit Row Height
Column Width...
AutoFit Column Width
Default Width...
Visibility
Hide & Unhide
Organize Sheets
Rename Sheet
Move or Copy Sheet...
Tab Color
Protection
Protect Sheet...
Lock Cell
Format Cells...
Hide Rows
Hide Columns
Hide Sheet
Unhide Rows
Unhide Columns
Unhide Sheet...
AC
AD

Scenario-Based Task: Hiding and Un-hiding Columns

A school secretary has created a spreadsheet to record learners' details and performance. The spreadsheet contains the following columns:

- Name
- Class
- Gender
- Test Score
- Comments (teacher remarks)
- Fee Balance

During a class presentation to learners, the teacher only wants them to see:

- Name
- Class
- Test Score

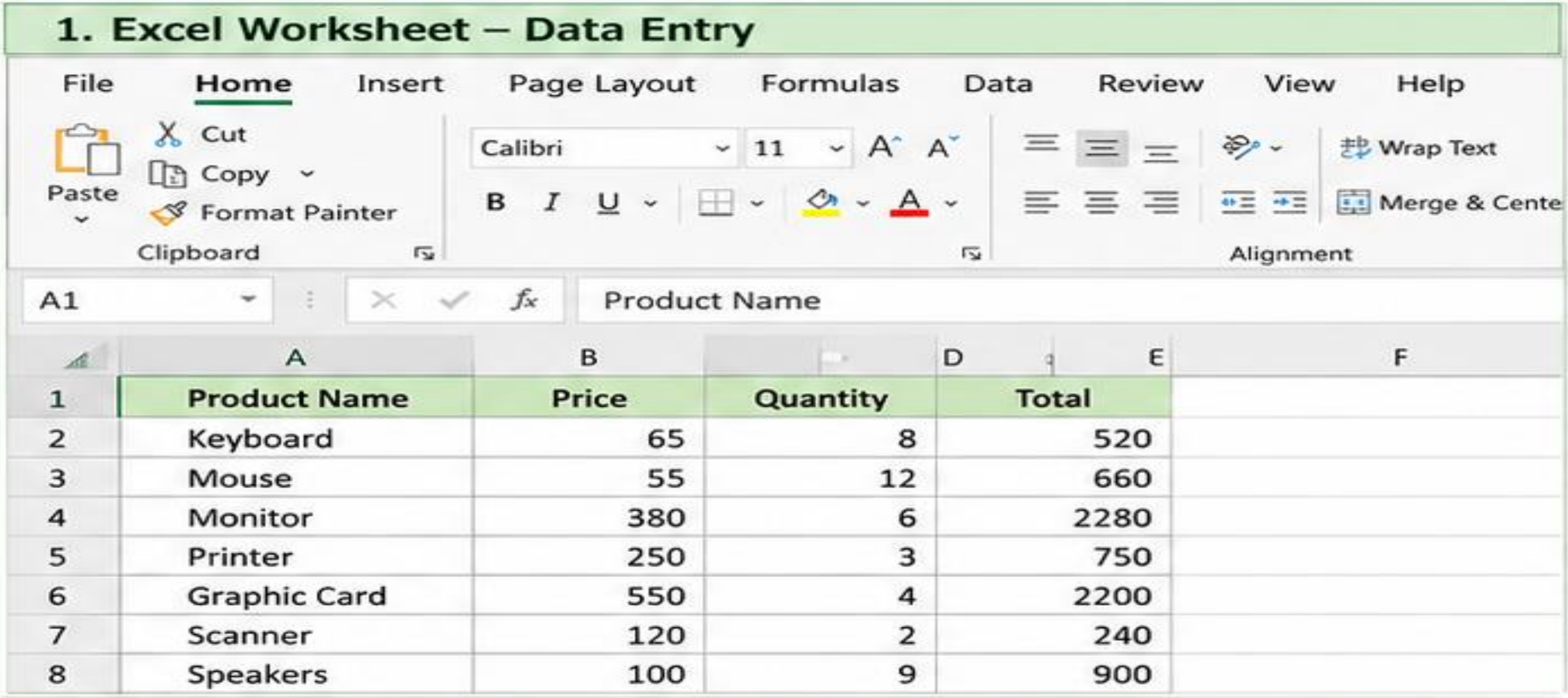
The columns **Gender**, **Comments, and Fee Balance** should not be visible during the presentation.

Tasks

1. Identify the columns that should be hidden.
2. Explain why those columns should be hidden during the presentation.
3. Describe the steps the teacher should follow to hide the selected columns in a spreadsheet.
4. After the presentation, explain how the hidden columns can be made visible again.
5. State one advantage of hiding columns instead of deleting them.

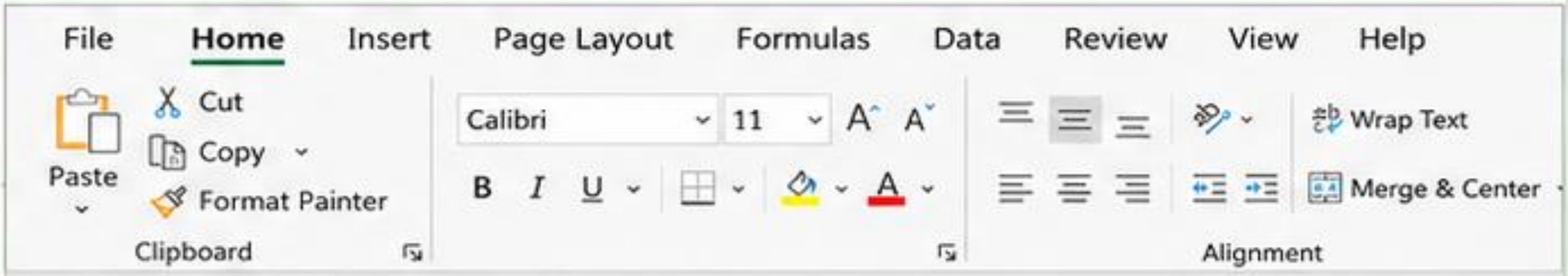
Activity – Work in Pairs

1. Excel Worksheet – Data Entry



	A	B	D	E	F
1	Product Name	Price	Quantity	Total	
2	Keyboard	65	8	520	
3	Mouse	55	12	660	
4	Monitor	380	6	2280	
5	Printer	250	3	750	
6	Graphic Card	550	4	2200	
7	Scanner	120	2	240	
8	Speakers	100	9	900	

2. Formulas in Excel



D2 | =B2*C2

	A	B	C	D	E	F
1	**Product Name**	**Price**	**Quantity**	**Total**		
2	Keyboard	65	8	=B2*C2		
3	Mouse	55	12	=B3*C3		
4	Monitor	380	6	=B4*C4		
5	Printer	250	3	=B5*C5		
6	Graphic Card	550	4	=B6*C6		
7	Scanner	120	2	=B7*C7		
8	Speakers	100	9	=B8*C8		

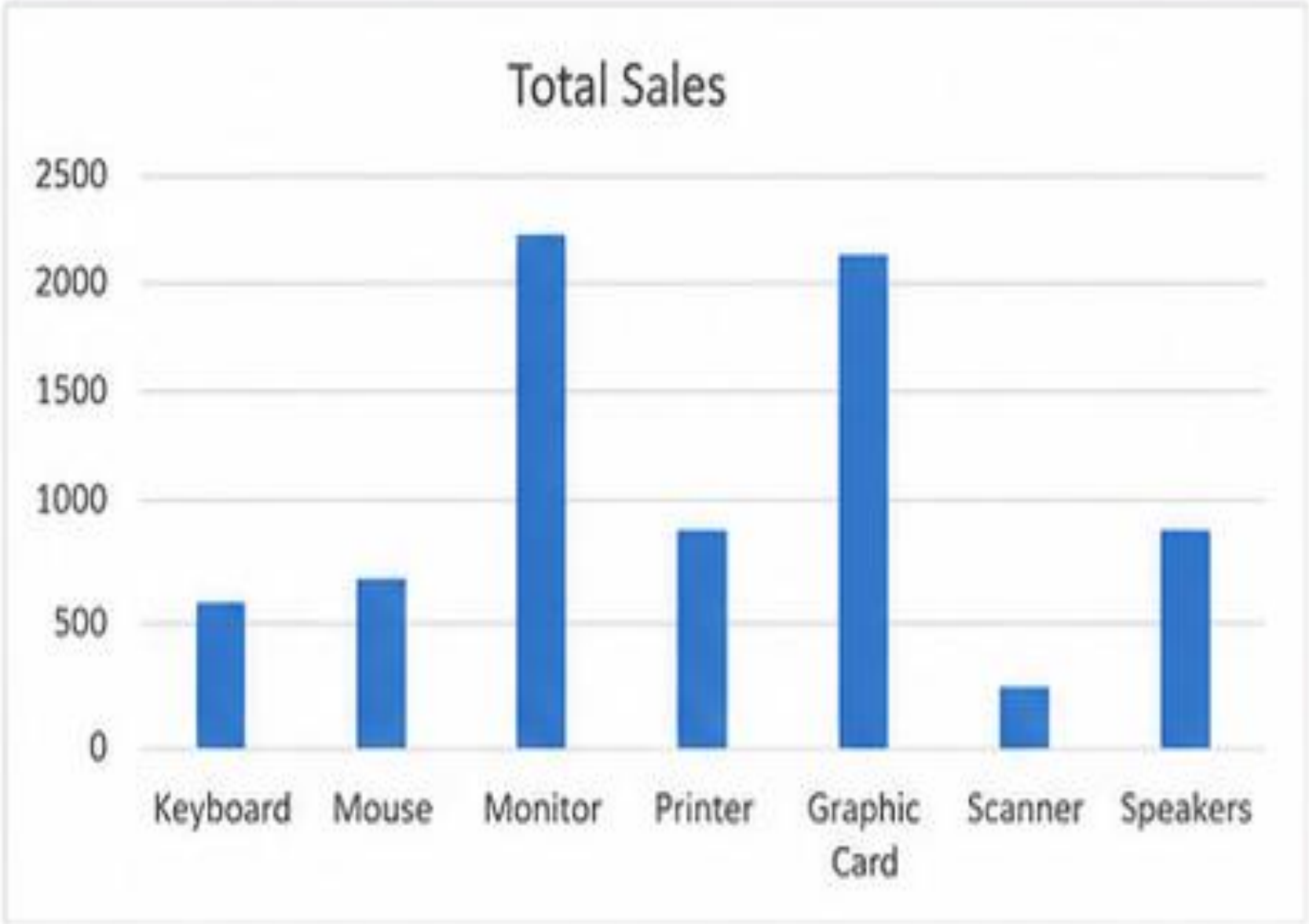
3. Using SUM – Total Calculation

D9 | =SUM(D2:D8)

	A	B	C	D	E	F
1	**Product Name**	**Price**	**Quantity**	**Total**		
2	Keyboard	65	8	520		
3	Mouse	55	12	660		
4	Monitor	380	6	2280		
5	Printer	250	3	750		
6	Graphic Card	550	4	2200		
7	Scanner	120	2	240		
8	Speakers	100	9	900		
9	**Grand Total**			=SUM(D2:D8)		

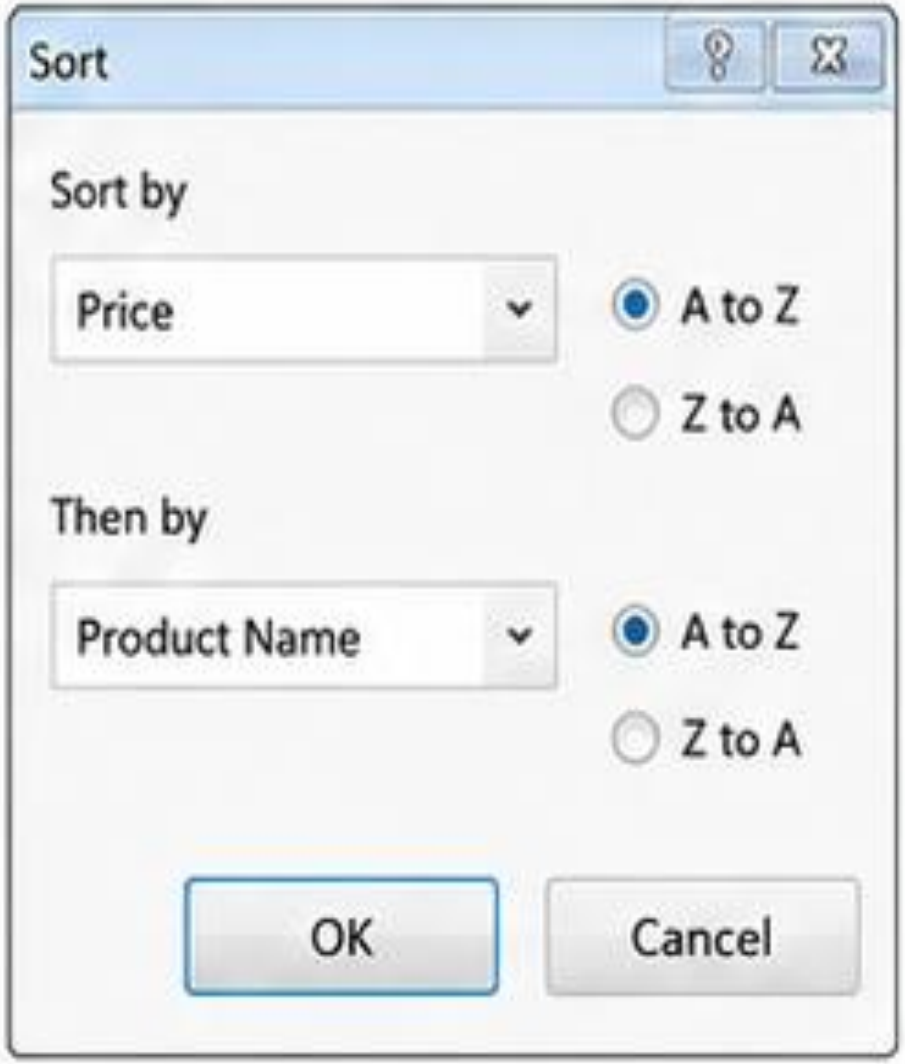
4. Chart in Excel – Column Chart

	A	D
1	**Product Name**	**Total**
2	Keyboard	520
3	Mouse	660
4	Monitor	2280
5	Printer	750
6	Graphic Card	2200
7	Scanner	240
8	Speakers	900



5. Sorting Data

	A	C	C	D
1	**Product Name**	**Price**	**Quantity**	**Total**
2	Scanner	120	2	240
3	Speakers	100	9	900
4	Mouse	55	12	660
5	Keyboard	65	8	520
6	Printer	250	3	750
7	Monitor	380	6	2280
8	Graphic Card	550	4	2200



6. Conditional Formatting – Highlight High Values

	A	C	C	D
1	Product Name	Price	Quantity	Total
2	Keyboard	65	8	520
3	Mouse	55	12	660
4	Monitor	380	6	2280
5	Printer	250	3	750
6	Graphic Card	550	4	2200
7	Scanner	120	2	240
8	Speakers	100	9	900



End of Topic
- Summative Assessment

PART 1: Spreadsheet Basics and Data Entry (10 Marks)

Question 1: School Club Records (5 Marks)

The Form 1 ICT Club wants to keep records of members who paid contributions for a school trip.

You are asked to create a spreadsheet with the following headings:

Name	Class	Amount Paid	Status

Task

a. Enter at least five learners' records into the spreadsheet. (2 marks)

b. Sort the data alphabetically according to learners' names. (1 mark)

c. Explain why sorting data is important in a spreadsheet. (2 marks)

Question 2: Organizing Market Sales Data (5 Marks)

A market trader recorded the following sales:

Item	Quantity Sold
Tomatoes	30
Onions	15
Cabbages	20

Tasks

a. Enter the data into a spreadsheet. (2 marks)
b. Apply bold formatting to the headings. (1 mark)
c. Insert borders around the table. (1 mark)
d. Use colours to highlight the heading row. (1 mark)

PART 2: Formatting and Spreadsheet Management (10 Marks)

Question 3: School Event Budget Formatting (5 Marks)

Your school is preparing for a JETS day event. Thomas, a learner created a spreadsheet showing expenses but it is difficult to read.

Tasks

a. State two reasons why spreadsheet formatting is important. (2 marks)

b. Describe how you would:

- apply bold text,
- insert borders,
- and use colours in the spreadsheet. (3 marks)

Question 4: Hidden Information in a Spreadsheet (5 Marks)

The Examination Council of Zambia (ECZ) hid some rows and columns containing examination marks while discussing results with the media.

Tasks

a. Explain the difference between hiding and deleting rows or columns. (2 marks)

b. Describe the steps used to:

- hide a row,
- un-hide a row,
- hide a column,
- and un-hide a column. (3 marks)

PART 3: Budget Creation and Functions (20 Marks)

Question 5: Family Monthly Budget (10 Marks)

A family recorded the following monthly expenses:

Item	Amount (K)
Food	2500
Transport	900
Electricity	600
Water	300
School Fees	1800

Tasks

a. Enter the data into a spreadsheet. (2 marks)
b. Use a spreadsheet formula to calculate the TOTAL expenses using the SUM function. (2 marks)
c. Find the HIGHEST expense using the MAX function. (1 mark)
d. Find the LOWEST expense using the MIN function. (1 mark)
e. Calculate the AVERAGE expense using the AVERAGE function. (2 marks)
f. Explain one advantage of using spreadsheets when preparing budgets. (2 marks)

Question 6: School Shop Stock Analysis (10 Marks)

Mr. Chumba, a tuck-shop teacher recorded the following items sold in one day:

Item	Quantity Sold
Pens	20
Books	15
Rulers	10
Pens	12
Books	8

Tasks

a. Use the COUNT function to count cells containing numbers. (2 marks)

b. Use the COUNTA function to count all non-empty cells in the item column. (2 marks)

c. Use the SUMIF function to calculate the total number of Pens sold. (3 marks)

You may use:

Pens Sold=20+12

d. Explain one situation where the SUMIF function is useful in real life. (3 marks)

PART 4: Practical Scenario Questions (10 Marks)

Question 7: Community Youth Project (5 Marks)

A youth group in your community collected CDF money to buy football jerseys. The chairperson wants the records to be neat and easy to understand.

Tasks

Describe how you would use a spreadsheet to:

- enter the data,
- format the headings,
- apply borders and colours,
- sort members' names,
- and calculate total contributions. (5 marks)

Question 8: Spreadsheet Features (5 Marks)

A Grade 11 learner opened a spreadsheet program to prepare class test results but could not identify some parts of the spreadsheet window.

Tasks

a. Define the following terms:

- Row
- Column
- Cell – (3 marks)

b. State any two features found in a spreadsheet window and explain their uses. (2 marks)

Next topic:

2.2 Productivity Tools

2.2.3 MS PowerPoint

2.2.3.1 Create presentations

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